# Loan Impairment Calculator

LIC in a BOX – User Manual

August 2017

Building a better working world

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# 1 Introduction

This document is a basic user guide for the Loan Impairment Calculator in a Box ("LIC" or "the tool" or "the application") tool. Its purpose is to provide the users with a quick navigation of the tool.

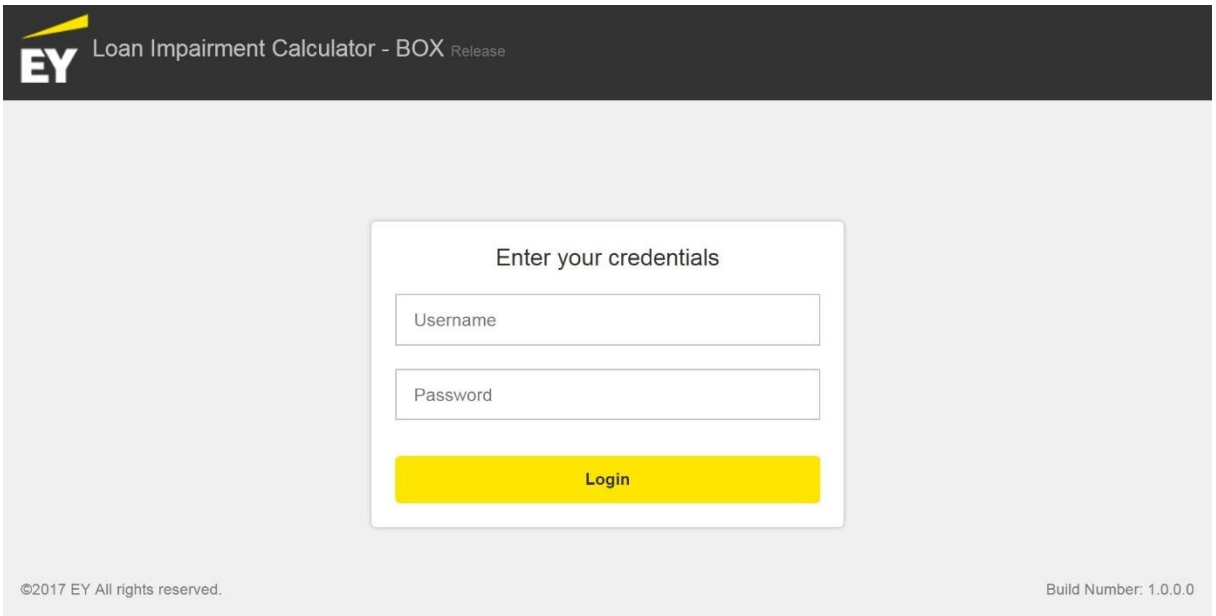
Methodological aspects are not covered in this document.

*The Loan Impairment Calculator is provided to the Bank for internal use only and must not be distributed to any third party without an explicit written permission of EY. The Bank accepts full responsibility for the security of the tool and the prevention of its unauthorized usage.*

# 2 Home page, data and run management

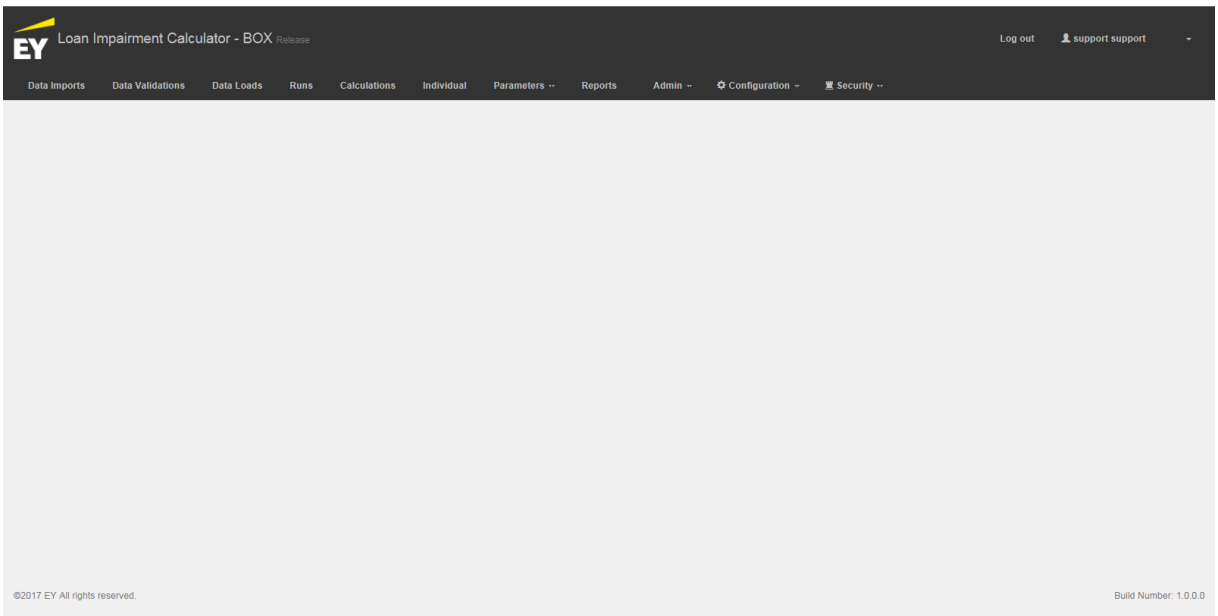
This chapter provides a description of the home page, data import, data load and run management features of the tool.

Before using the tool, the user has to enter a valid username and password in the login dialog box provided by EY support team. Once the Active Directory (AD) authentication is activated, there will be no login dialog and the user will be authorized based on his/her AD group membership.




## 2.1 Home page

The home page, as well as all the other application screens, displays the application's main menu which is accessible from the bar at the top.



From the home page it is possible to access:

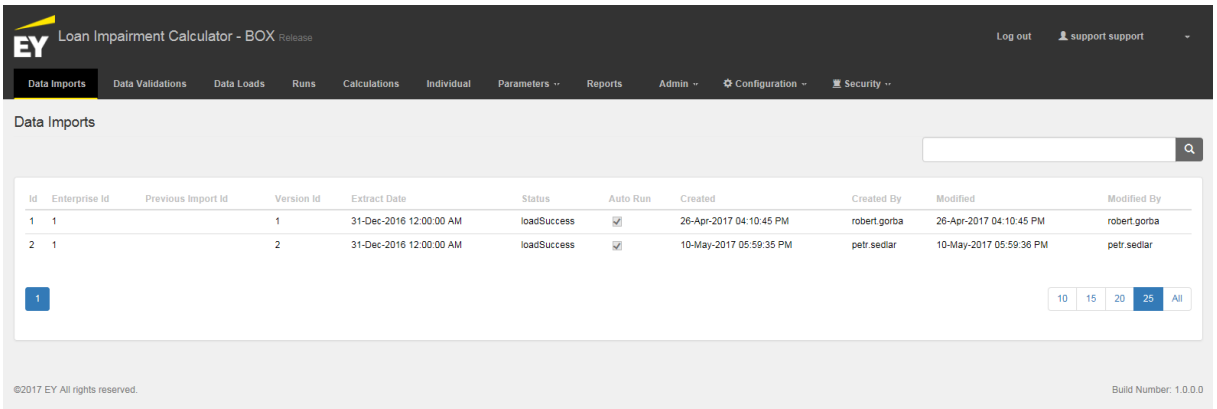
- **Data imports:** In this section, the user can find the list of available data imports.
- **Data Validations:** This section contains results of data quality checks.
- **Data loads**: This section contains the list of available data loads.
- **Runs:** This section provides an overview of all the previous calculations and allows to create new calculations.
- **Calculations:** In this section, the user can view the calculation process of a selected run.
- **Individual:** This choice displays the Individual assessment module including the list of both manually and automatically selected clients. Clients can be accessed and reviewed in this section.
- **Parameters:** This section allows the user to set the calculation parameters.
- **Reports:** This section allows the user to manage reports.
- **Admin:** This section lets the user set up system settings and to manage portfolios.
- **Security:** In this section, the authorized user may adjust settings concerning roles and users.

<*Note:* The order of the menu items and the items visible might be different for each user based on the current application configuration and the setup for individual user roles.>

## 2.2 Data imports

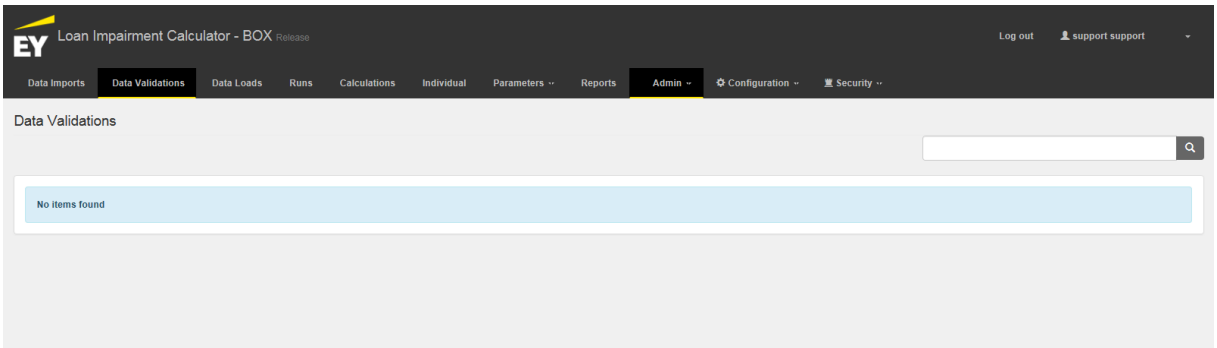
This section shows all the existing data imports from the source system. The table below provides an overview of the existing data imports and contains

- *ID* (import ID),
- *Extract Date* (extract value date),
- *Version ID* (version ID of loaded input data assigned during the import&load process – please note that this relates to *Data Load* when creating a run manually),
- *Status* (status of the import&load process),
- *Auto Run* (information if  Auto Run was selected during the import&load process),
- *Created* (time at which the *Start import* command was executed) and
- Other details regarding each data import.



## 2.3 Data validations

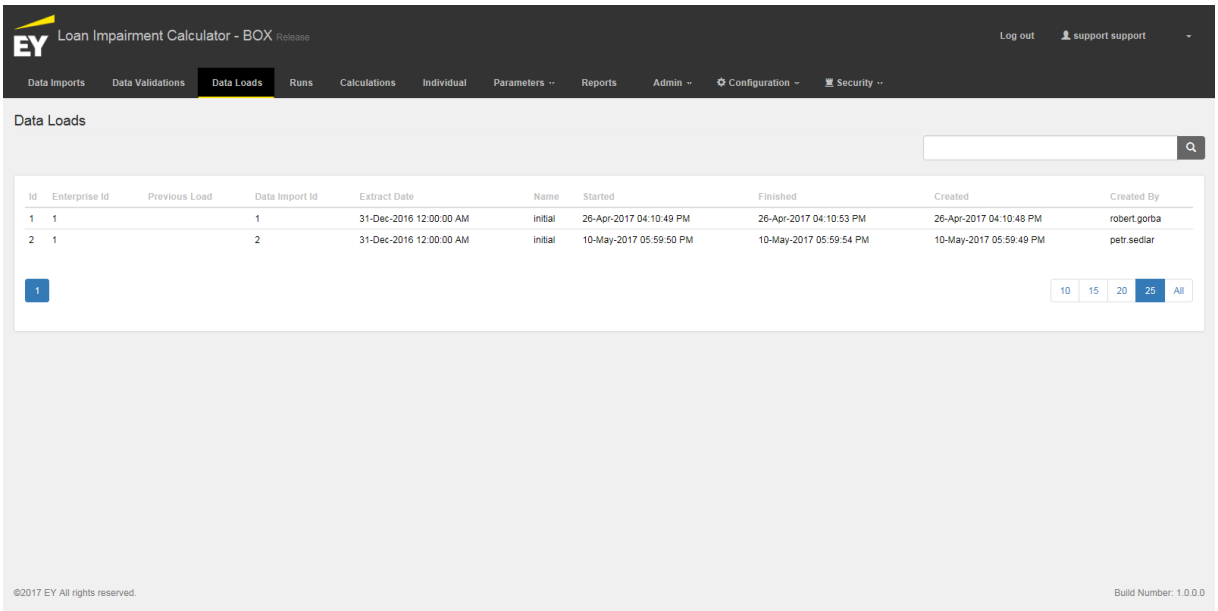
This section is under construction and will be updated in further releases.




## 2.4 Data loads

This section enables the user to view the list of data loads. The table presents an overview of the existing loads with basic information about each load:

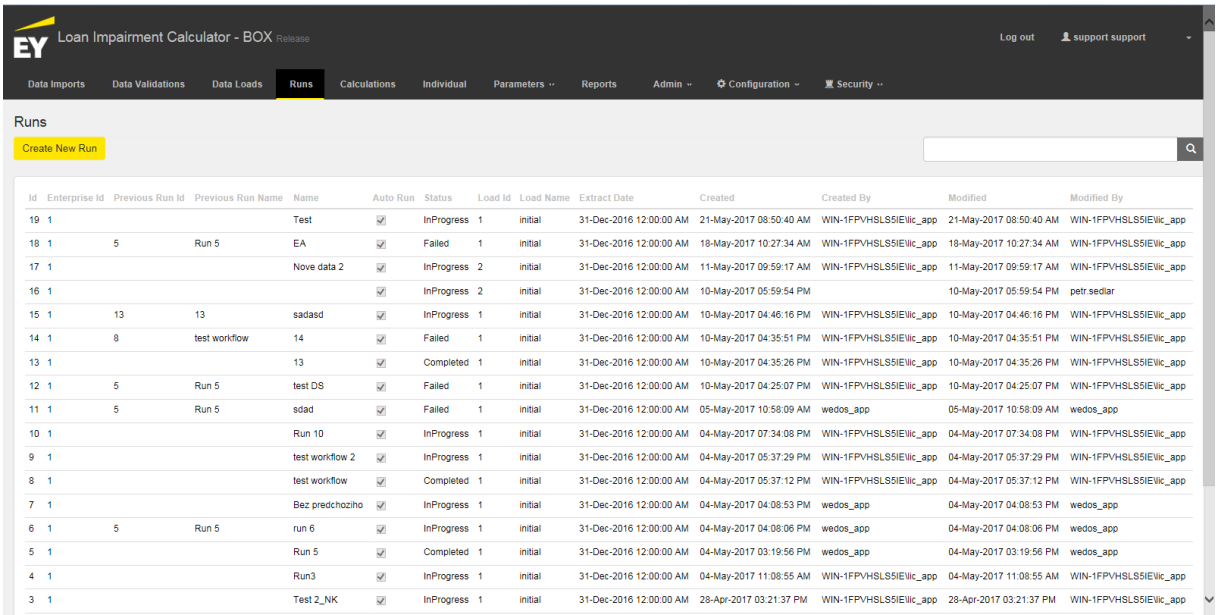
- *ID* (Version ID from Data Imports page, or in other words, internal Version ID of loaded data),
- *Previous Load* (previous load Version ID),
- *Data Import ID* (relevant import ID from the load&import process),
- *Extract Date*,
- *Name*,
- *Started*,
- *Finished*,
- *Created* (time at which the import&load process initialized physical data load from DataStaging to LIC input scheme) and
- *Created By*.



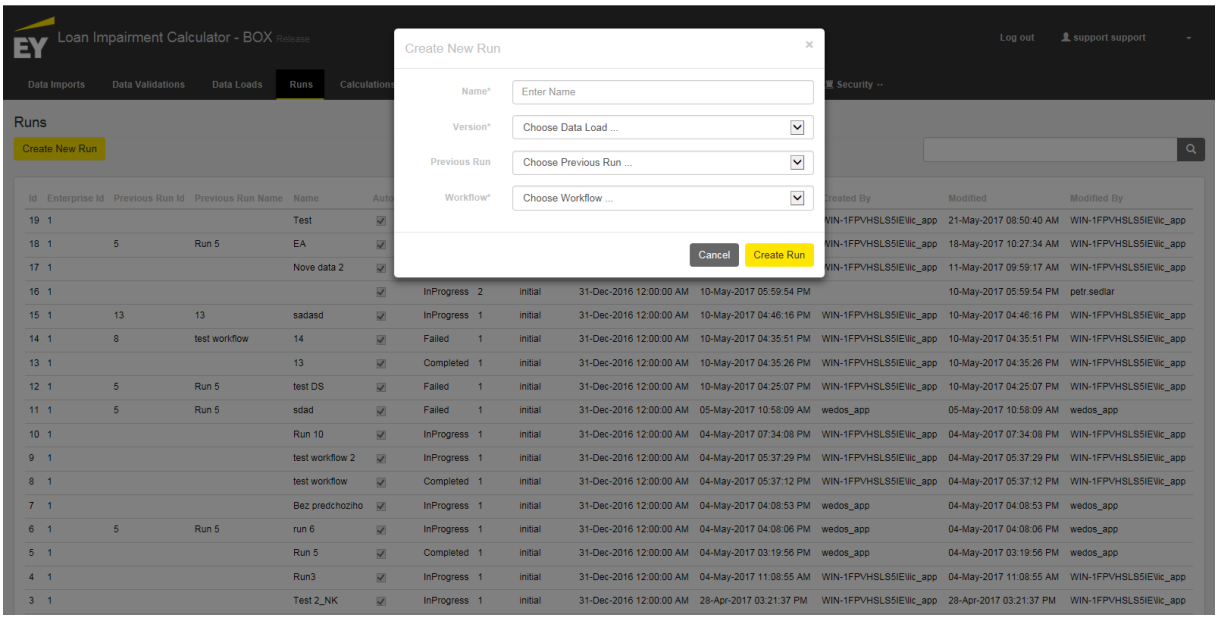
# 2.5 Runs

This section presents an overview of all the previous calculation runs and enables the user to create new runs.

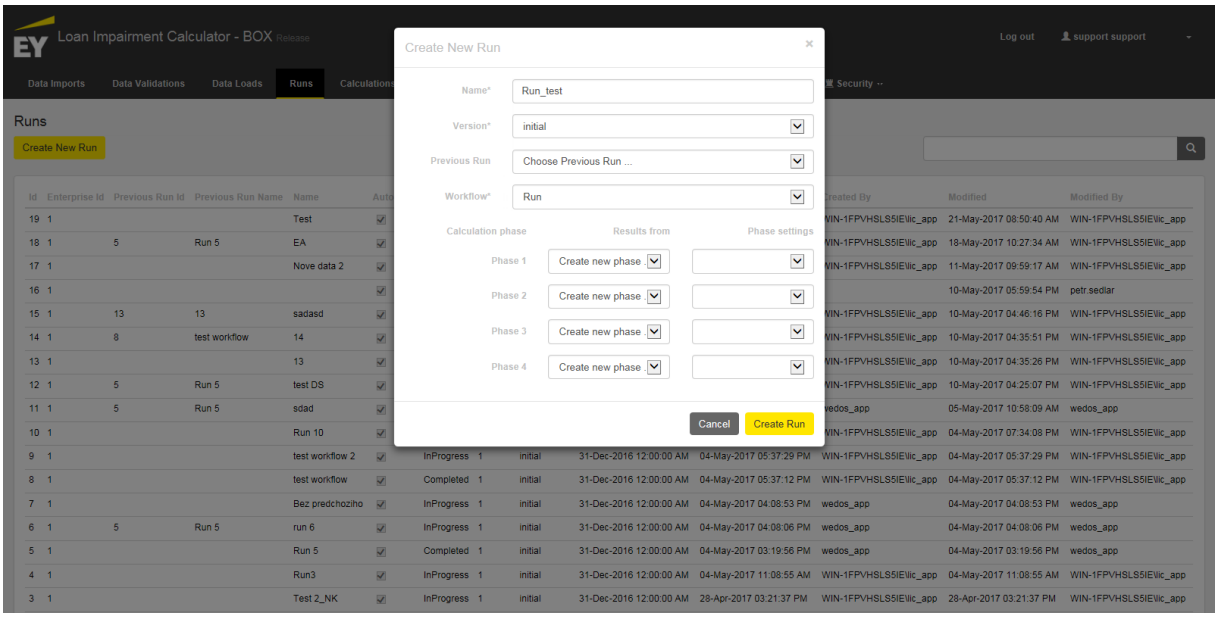
The default page view is captured in the following screenshot.




By clicking on the *Create New Run* button the user is requested to enter specifications for the new run to be created.




In order to create a new run, the user must input a *Name* for the run, select the intended *Version* of data load (drop down menu), specify whether there is any *Previous Run* (drop down menu), and select the *Workflow* to be applied in the run. There is only one type of run workflow available in the current version of the tool. In future versions, for each phase within the workflow, the user will be able to select a phase from which the data should be applied (a new phase can be created or results from other runs can be used) and define the *Phase settings* as shown in the screen below. At the moment, the *Phase* section should be left as it is.

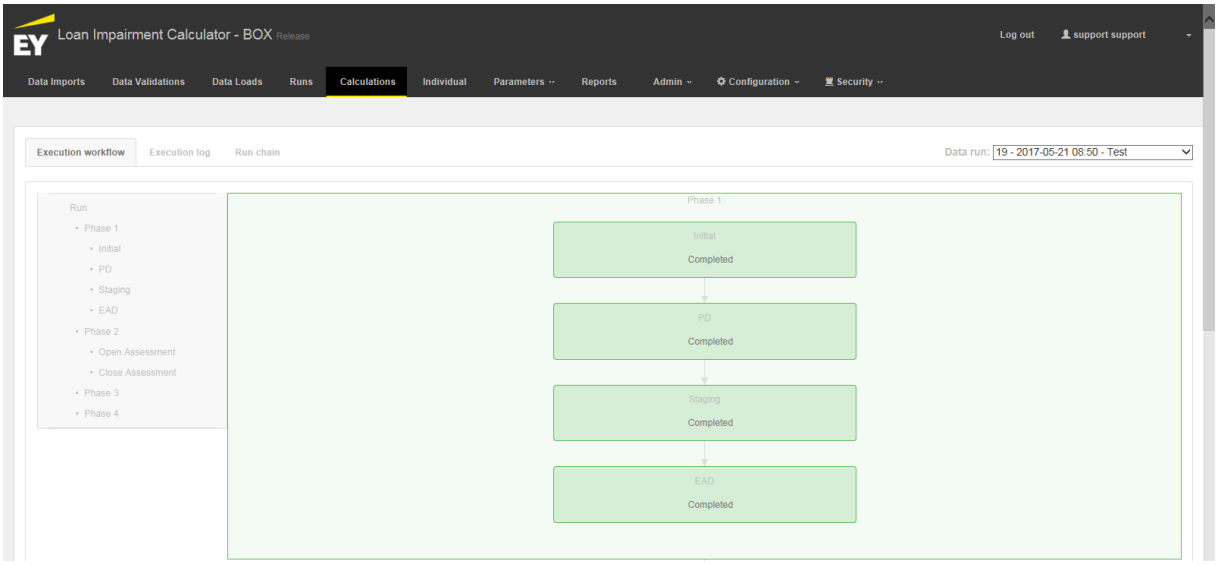



After filling out the required information the run setup can be confirmed by clicking on the *Create run* button.

# 3 Calculations

One of the features of the LIC tool is a calculation process tree that displays the current state of calculations. In the process tree, every small box (e.g. Staging) represents a calculation step and every large box (e.g. Phase 1) represents a calculation phase.

After creating a new run, the calculation starts automatically and its progress is shown in the process tree: when a box has a grey outline, it is waiting for its prerequisites to be completed. Green outline means that the process has already been completed. Orange indicates that the process has failed.




Generally, the process consists of the following main phases:

- **Phase 1** – Probability of Default (PD), Exposure at Default (EAD) and Staging
- **Phase 2** – Opening and closing individual assessment
- **Phase 3** – Credit Loss Allowance (CLA) Calculation
- **Phase 4** – Output data generation

Each phase contains workflow parts detailed in the table below.

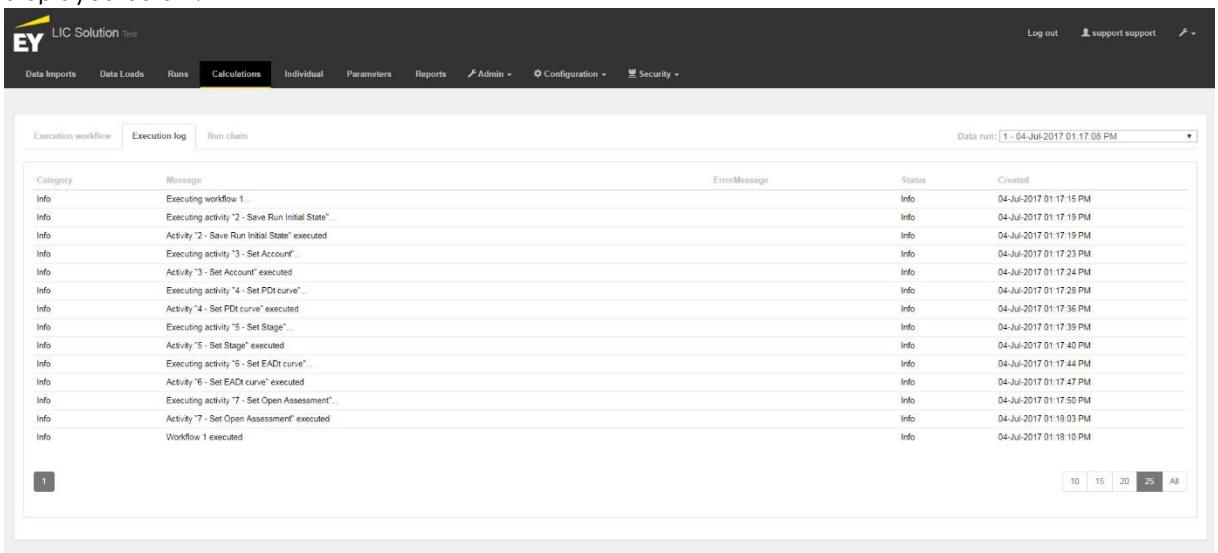
**Table 1: Calculation workflow**

| Phase | Area |
|---|---|
| Phase 1 | Initial |
| | PD |
| | Staging |
| | EAD |
| Phase 2 | Open Individual Assessment |
| | Close Individual Assessment |
| Phase 3 | CLA Calculation |
| Phase 4 | Fill Output |

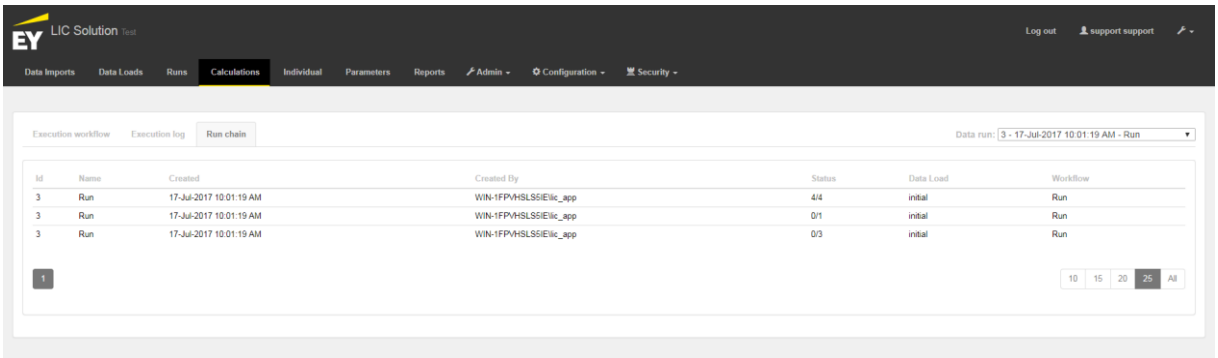
All calculation runs (even those that were not completed) are saved and their overview is provided on the *Runs* page.

Apart from the *Execution tab*, there are two more tabs on the page: *Execution log* and *Run chain*.

The *Execution Log* provides more detailed information about the run procedure. Its environment is displayed below.




*Run chain* is displayed in the picture below.

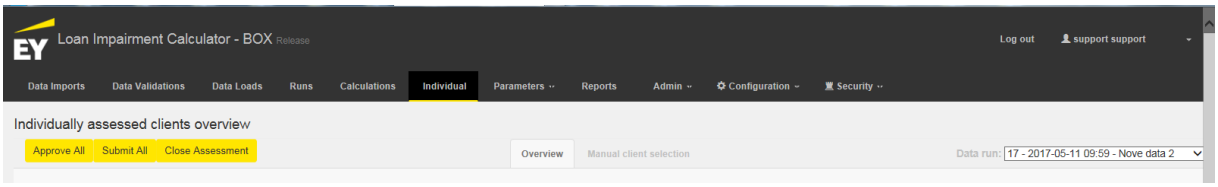


# 4 Individual assessment

The individual allowance calculation is performed under the *Individual* module, where an overview of the individually assessed clients is presented.

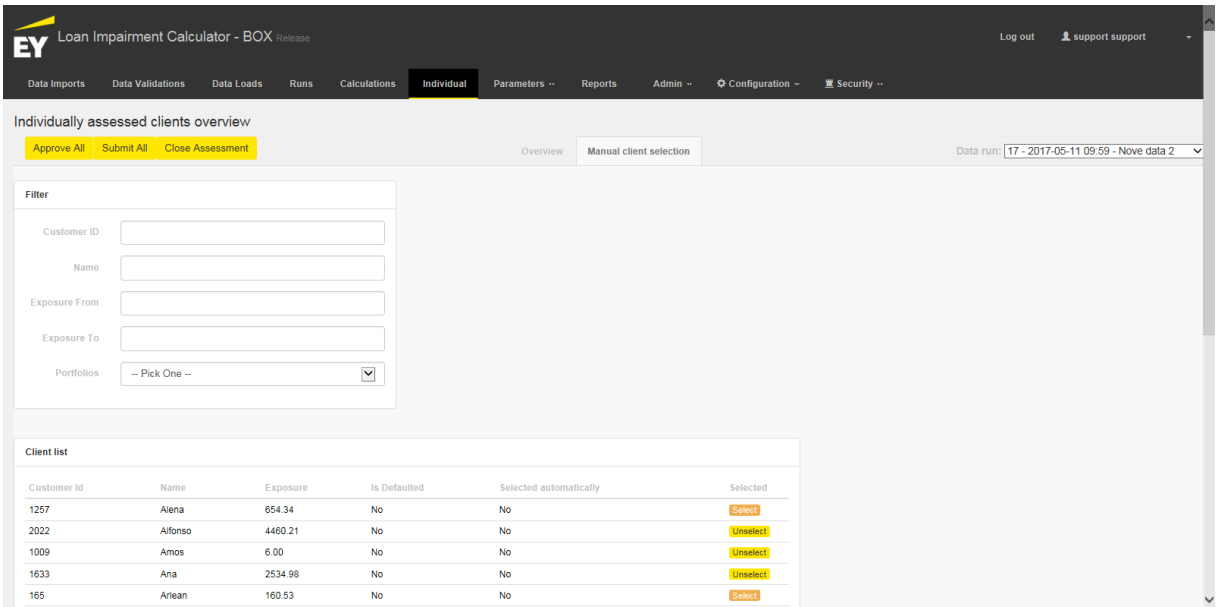
Initially, there is a set of clients automatically selected during the selection for the individual assessment process (criteria for automatic selection can be configured in *Parameters* > *Individual Methods*, see section 5.3). Moreover, in the *Manual client selection* tab, the user has an option to choose additional clients to be assessed individually.

Reference date of calculation is set during the process of setting up a new run. It is always visible in the upper right corner in *Data run* drop-down menu.




## 4.1 Selection of customers for individual assessment

In this section, the user may manually select clients for individual assessment. These manually selected clients will appear in the *Individual* section of the tool along with the clients that met the automatic selection criteria.




The picture above shows the form for manual selection of clients.

The top section of the page provides a filter which allows the user to search for clients by ID, name, portfolio, or by a targeted exposure.

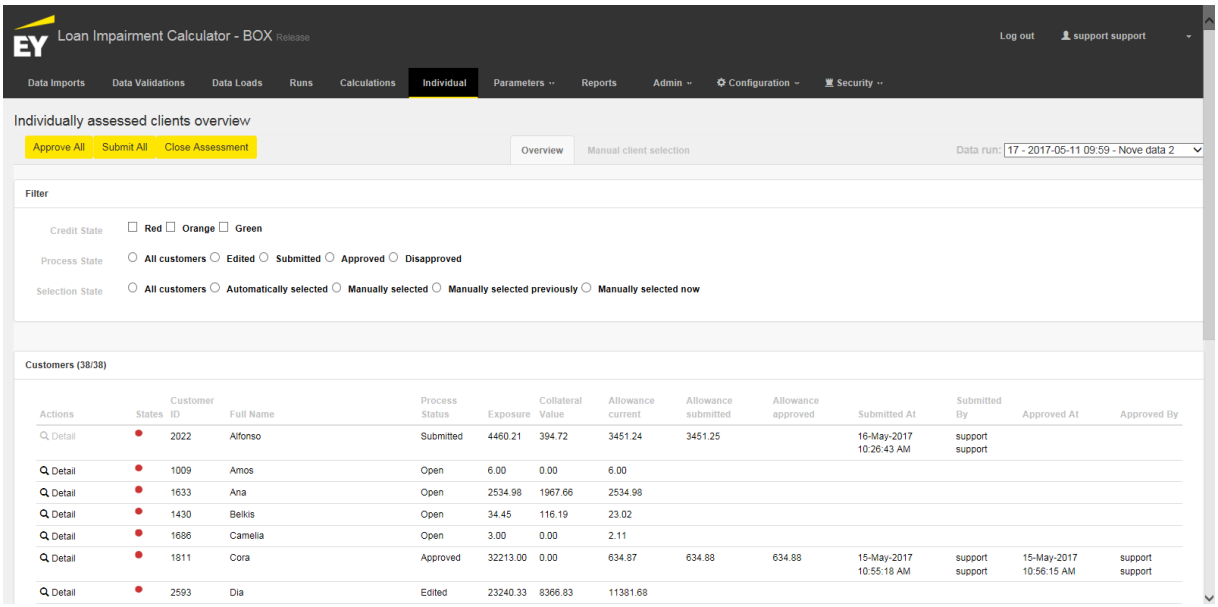
It is possible to only fill out the *Exposure From* field to find clients with exposure above a certain level, or to only fill out the *Exposure To* field to find clients with exposure below a certain level. It is also possible to fill out both the *Exposure From* and the *Exposure To* fields to find clients with exposure in a particular interval.

The list of page numbers at the bottom of the screen allows the user to move from one page to another if the client list exceeds one page. Clients can be selected using the *Select* button. Any changes will be automatically saved.

If a client is selected for individual assessment for a reason other than automatic selection in the current run (i.e. manual selection in previous runs or fulfilment of automatic selection criteria in previous runs that no longer hold), they can be manually unselected. Such reverse selection can be executed using the *Unselect* button; any changes are saved automatically.

# 4.2 Individual assessment client overview

When the user completes the manual selection of clients, the *Overview* page displays a complete list of clients to be individually assessed.




The clients are divided into 3 states – red, orange and green based on review prioritization criteria. Criteria should be specified in *Parameters* > *Individual colors* (more details in section 5.4). Individual assessment can be closed only after all "red" clients have been assessed. In case some of the red segment clients are not assessed, LIC displays a warning message. Orange and green clients can be left unassessed. All clients need to be assessed at least at the very first month of their entry into the individual assessment.

The Filter tool above the table of customers allows the user to filter for:

- Clients belonging to a specific color segment (green, orange, red)
- Clients who are edited, submitted, not submitted, approved, or disapproved
- Automatically or manually selected clients

### Submit/Approve assessment

The *Detail* button in the leftmost column opens each client's individual *Customer assessment* page. At this page, the user can submit an assessment of a client. The user also has an option to submit all assessed clients by pressing the *Submit all* button on the *Individually assessed clients overview* page. Submissions must be then approved; approval is possible on the client level or using the *Approve all* button. The user has to be authorized to be able to submit or approve clients. .

### Close Assessment

Once the assessment is closed, no user can continue to work with the current run. Please do not press the *Close assessment* button unless all clients that should be assessed within the run are finalized.

## 4.3 Customer assessment

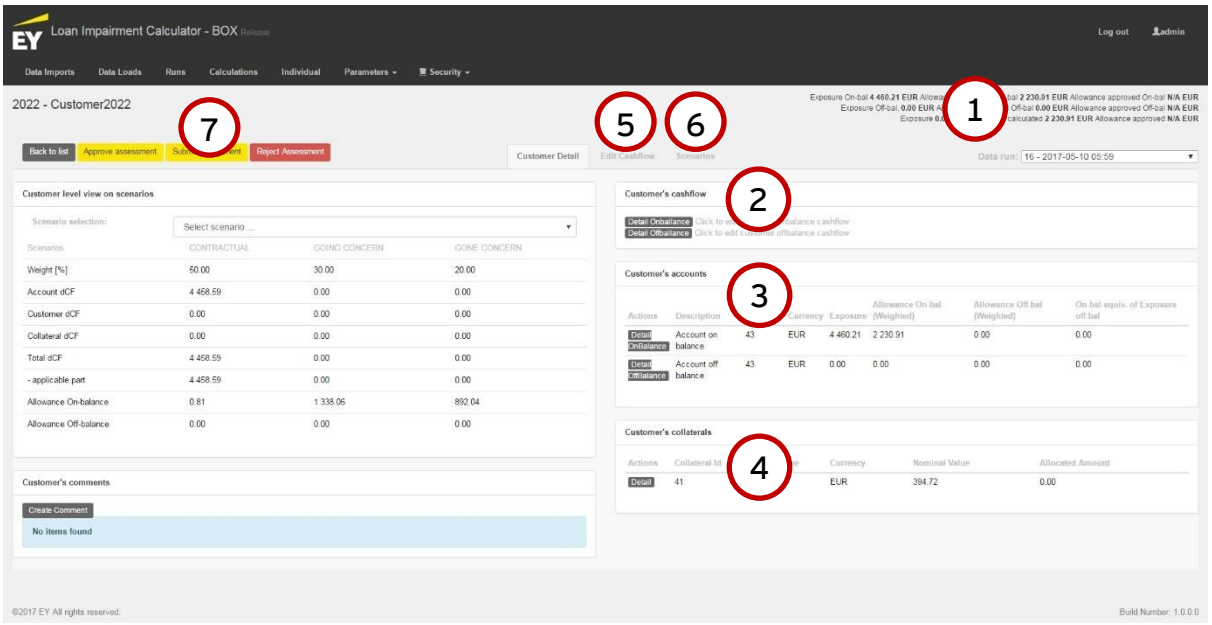
The *Customer assessment* page is accessible using the button *Detail* in the previously described page – *Individual assessment client overview*. The *Customer assessment* page is designed to provide a comprehensive review of a particular client.

Individual assessment screen works in the following two contexts:

- Selected scenario
- Selected cash-flow (CF) for assessment (by clicking on a particular *Detail* button).

This is because when the user opens a client page for assessment, they should first select the scenario they want to work with (or create a new one), and then select an appropriate entity (e.g. client CF, account CF or collateral CF).

The *Customer assessment* page, as described in the picture below, provides information about the following:



## 1- Client

The part related to the information about the client is visible in the upper right corner of the screen. This section displays the client identification (ID, name) on the left side of the screen, information about current exposure (on-balance and off-balance) and individual allowances on the right side of the screen.

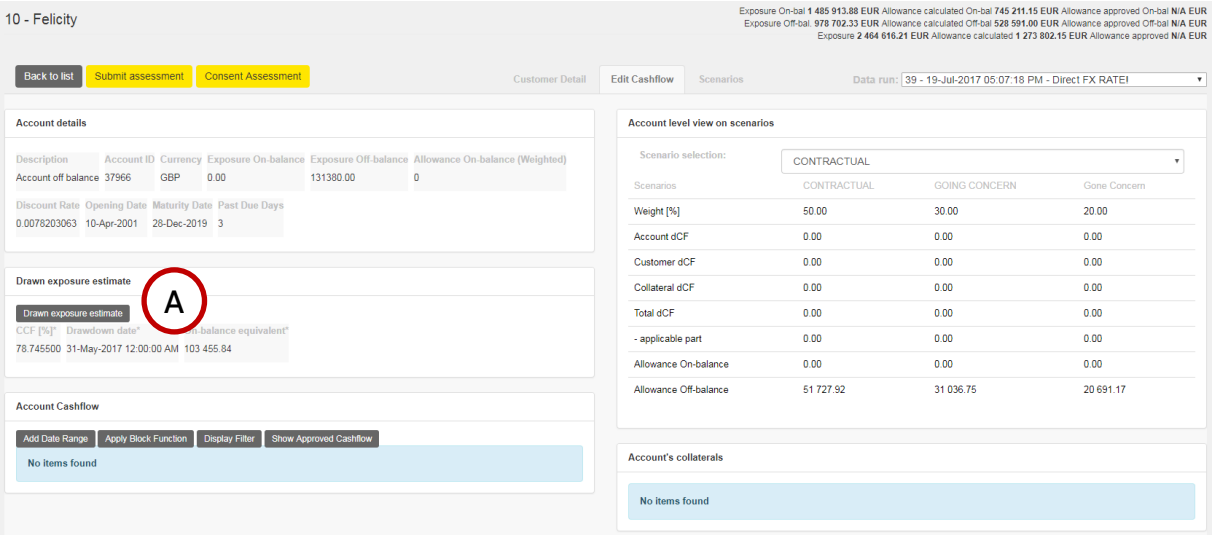
## 2 - Customer Cash flows

From this page, the user can access and modify cash flows on the customer level (for a selected scenario). In order to do so, the user should click on the *Detail* button. The tool will redirect the user to edit customer cash flows. By default, the client cash flows are allocated to the accounts based on the ratio of their exposure to the total exposure (both on- and off-balance). This allocation can be changed on ad-hoc basis (for more details see section 4.3.2). The user can provide a different allocation as percentage of the client cash flow assigned to a particular account based on his/her professional judgment.

## 3 - List of client's accounts

This sub-section presents an overview of all the client accounts – Description, IDs, currency, exposure, allowances on- and off-balance (weighted) and on-balance equivalent of exposure off-balance (Draw down estimate).

The details of the assessment of individual accounts can be accessed by clicking on the *Detail OnBalance/OffBalance* button. In the redirected screen, the following information is displayed for each account – description, label, currency, exposure on-balance, exposure off-balance, allowance weighted, discount rate, opening date, maturity date and past due days. In case of the *Detail OffBalance* page, the screen also contains the *Drawn exposure estimate* table. The button *Drawn exposure estimate*, marked as *(A)* in the picture below, allows the user to set the CCF coefficient value and particular drawdown date. The details of the screen are further discussed in section 4.3.1.

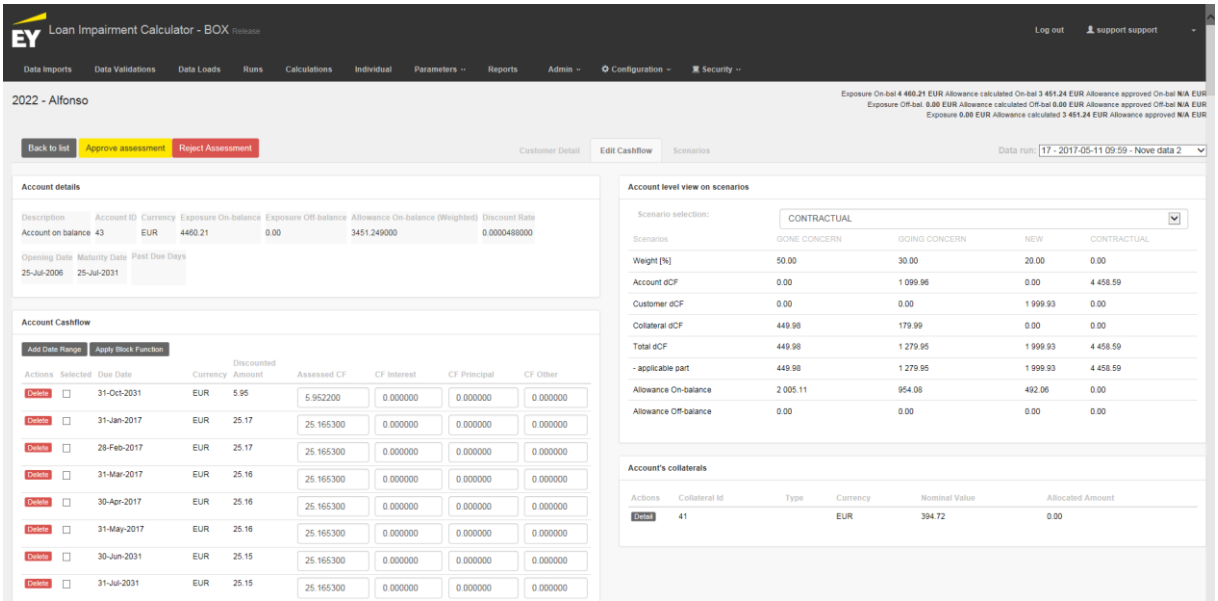



## 4 - List of client's collaterals

This section contains a table with the client's collaterals, their ID, type, values and currencies. Even collaterals with zero market value are included. When an account-level assessment is performed, only collaterals allocated to the particular account are displayed. Through this section, collateral details may be accessed by clicking on the *Detail* button (type, description, collateral cash flows).

## 5 - Cash flow assessment

As a part of the assessment of allowances, future cash flows for the account are estimated. The cash flows come from three sources: client's payment schedule, account-level payments and expected realization of collateral.

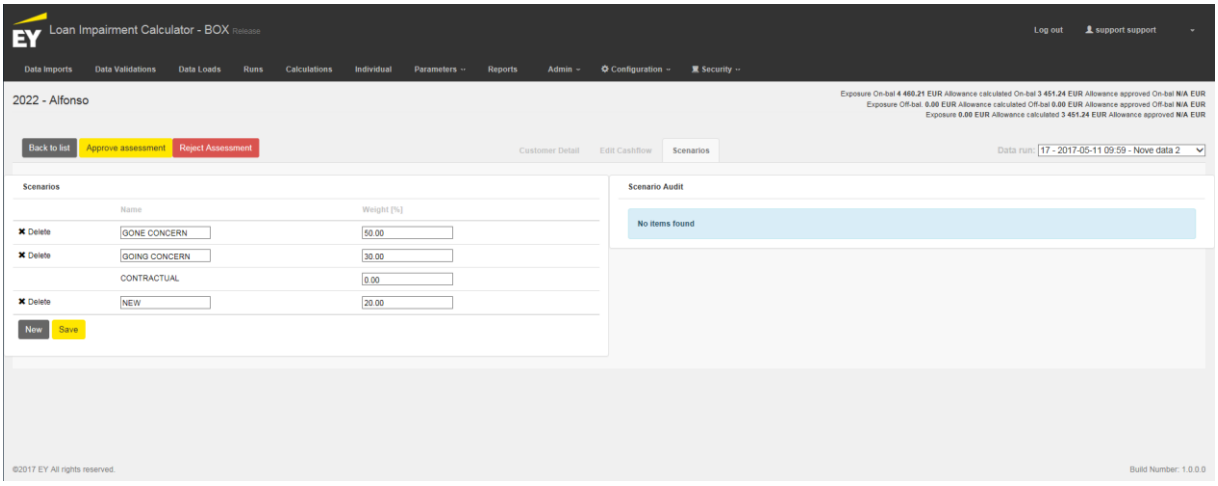



## 6 – Scenarios

This part of the page, accessible after the selection of a cash flow source, displays the scenarios of possible expected future CF. Along with the scenario name, it also shows the scenario weight (probability). The user selects the desired scenario to activate it. All actions in the bottom part of the screen are subsequently used as an input for the selected (active) scenario.

Aggregated discounted cash-flow (dCF) by type (customer dCF, account dCF, collateral dCF), the total sum of dCF and the applicable part of total dCF (the part of dCF which is relevant for allowance calculation) are displayed in the CF details on the scenario level.

The environment is displayed in the picture below.

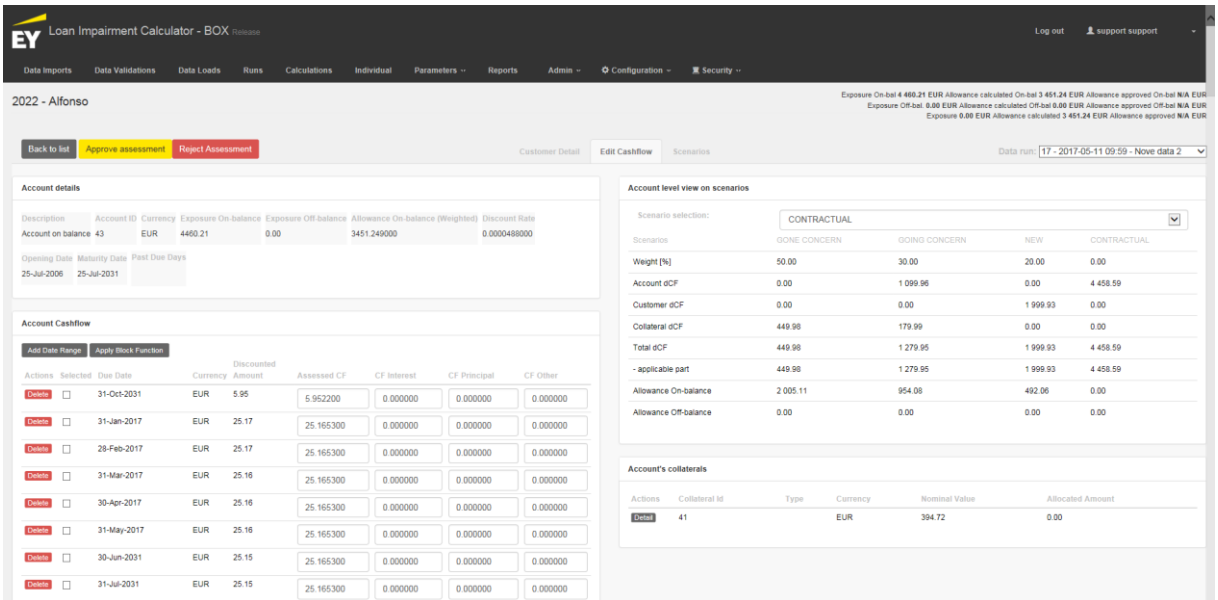


### 7 – Submit client

The base page for the assessed client also serves for assessment submission. This can be executed by clicking on the *Submit assessment* button. By doing so the user confirms that the client has been assessed. After this confirmation, the allowances are stored within the LIC. In case changes are needed once the assessment has been submitted, an authorized user may disapprove submission by clicking on the *Reject Assessment* button. After the changes are made the assessment should be submitted and consequently approved by clicking on the *Approve Assessment* button. The user can manage the assessments only if he/she is authorized to do so.

## 4.3.1 Account-level cash flows

The following picture presents an example of the account-level view, where cash flows for a particular account may be entered. This environment is accessible by clicking on the *Detail OnBalance/OffBalance* button in the accounts block (highlighted as *(3)* in the previous section of this document) on the base screen of customer assessment.




### Account cash flow table – overview

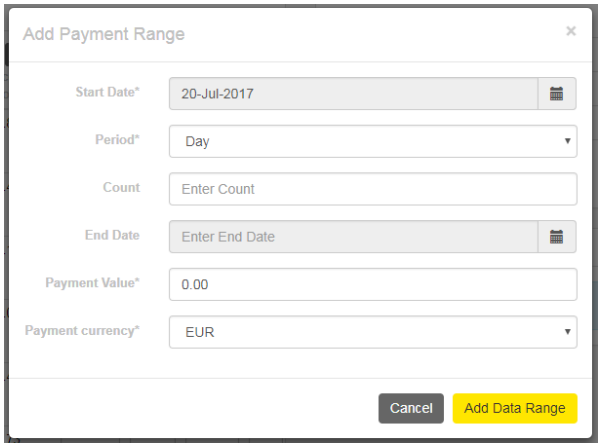
In the *Assessed CF* column of the account level cash flows the user can enter the expected payments either manually or by applying a function. In case of a manual edit it is required to use payment date(s) that are already in the table. Second option is to use *Add Date Range* or *Apply Block Function* functionalities (fully explained later in this section). Third possibility is to apply one of the many **predefined scenarios** (explained in section 4.3.3) that cover usual payment scenarios.

### Operations

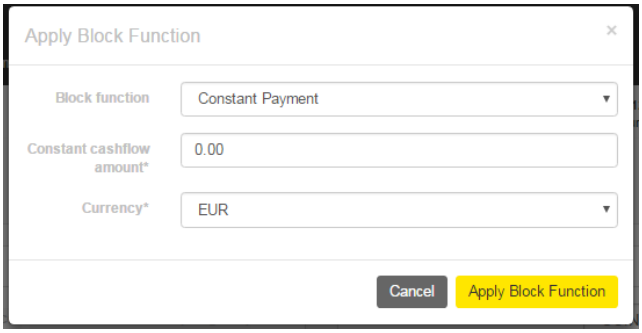
There are several ways to adjust the assessed cash flows. Apart from the manual editing of rows, these are:

- *Add Date Range*
  - This operation creates new rows in the table. It is intended to generate a number of dates by entering the *Start date*, *Period*, *Count*, *Principal value*, *Interest value* and *Payment currency* parameters. By clicking on *Add date range,* new rows are created. First, the user is asked to set the *Start Date* of the period at which the payments are expected. Second, the user sets the frequency of the payment by choosing the *Period* from the dropdown menu. Third, the user sets a number of payments in the period in

the row named *Count* to define the length of the payment range, or defines the *End Date*, at which the last payment is expected. Then the user sets the amount to be paid each period. Finally, the currency of payments is selected.




- *Apply Block function*
  - By applying a block function, the selected rows from a table will be adjusted. There are three functions implemented. Function is applied by clicking on the *Apply Block Function* button after the function parameters are inserted.




Description of functions available:

- *Constant payment*

  *Constant cash flow amount* filled in by the user is inserted into all the chosen payments in the calendar. If there is already a non-zero cash flow in the chosen payment, the constant payment is added to this value.
- *Percentage of contractual cash flow*

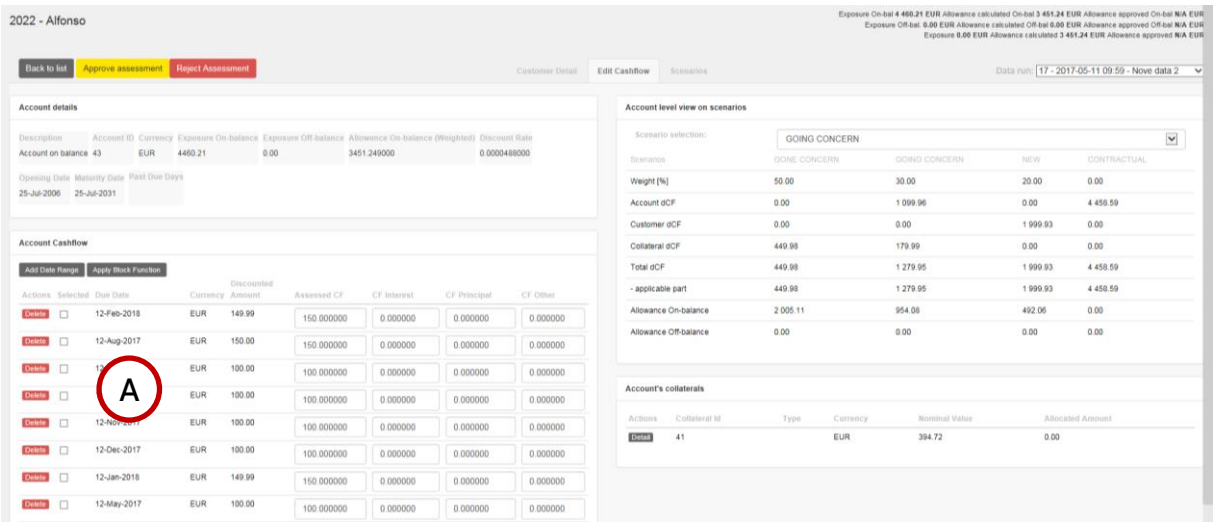
  Percentages of contractual payments (at the same date) are used for chosen payments in the calendar.
- *Delay payments*

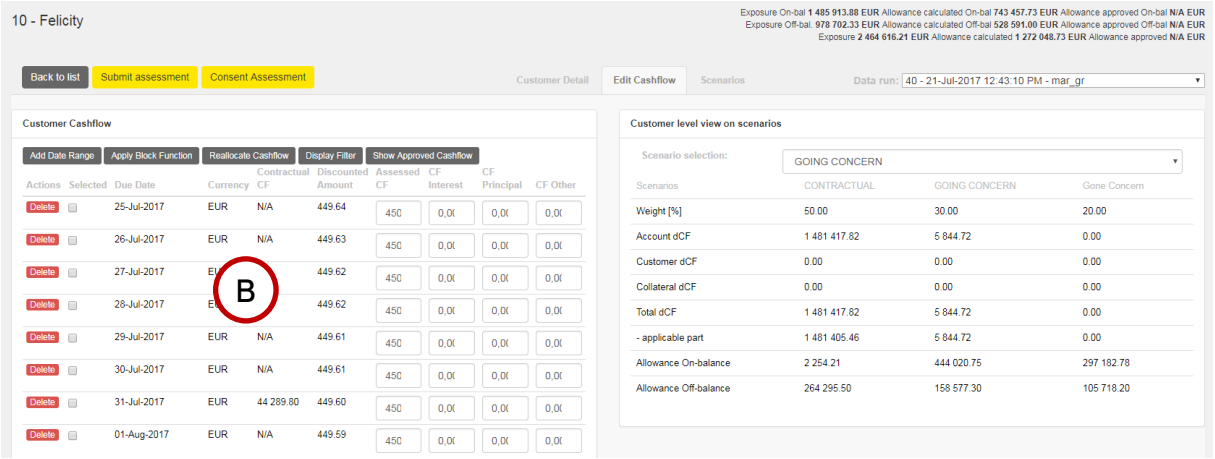
  Chosen payments are delayed by a period selected by the User.

## 4.3.2 Customer CF and Collateral CF Allocation

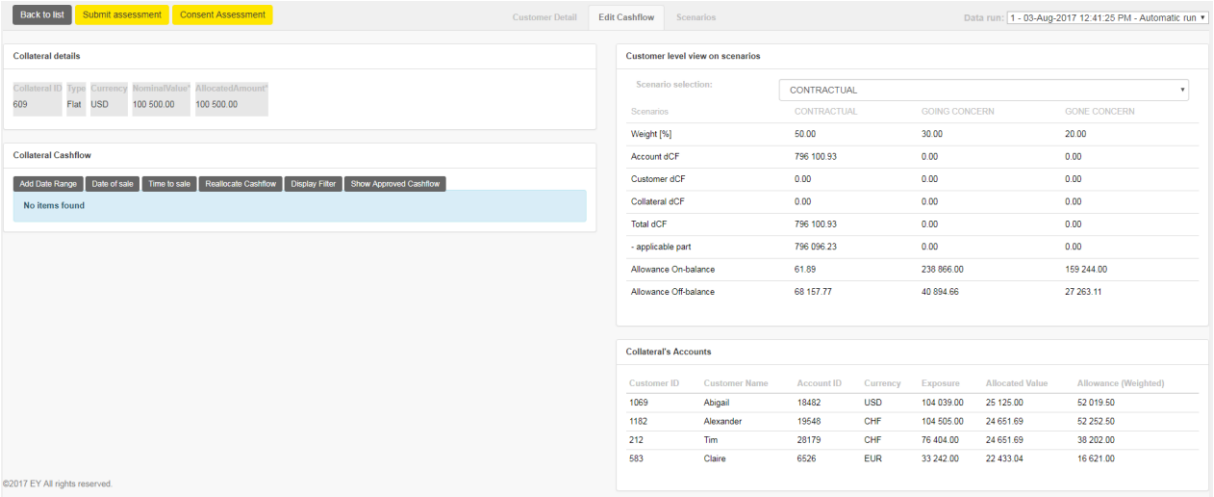
The customer CF and collateral CF have to be allocated to accounts. By default, the customer CF is allocated to accounts in the ratio of their exposures and the collateral CF is allocated to accounts by the Bank's input files.

Customer CF is edited using the *Add Date range* or *Apply Block Function* buttons, marked as (A) in the figure below. It can also be edited manually in each cell. Allocation of CF to particular accounts can be changed by using the *Reallocate Cash flow* function marked as *(B)* in the picture below.



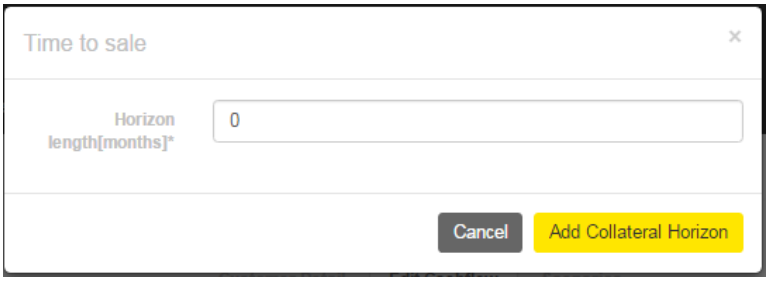

The following picture presents an overview of the collateral cash flow input:



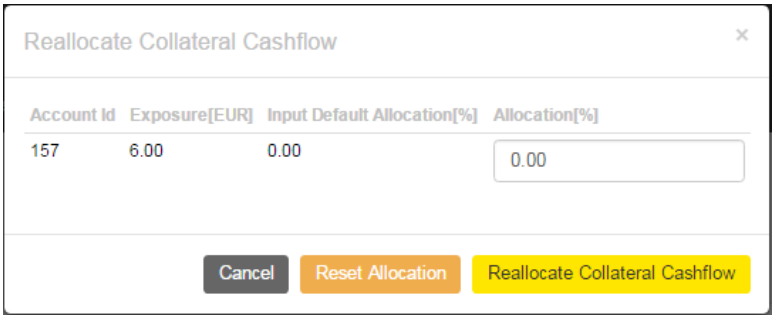

As in case of the account cash flows, the user may use the *Add Date Range* and *Apply Block Function* features. The functions that may be applied include constant payment, percentage of contractual CF and delay payments.

<***Note:*** if the *Delay block function* is used for collateral cash flows, in the next run the same delay period is applied on new dates changed since the previous run (if collateral has not been sold yet).>

Apart from these two features, the user can also add a collateral horizon *(Time to sale)* by inputting the expected collateral recovery horizon length in months. The entry must be confirmed using the *Add Collateral Horizon* button.




The *Reallocate CF* button provides a possibility to reallocate collateral cash flows among loans.



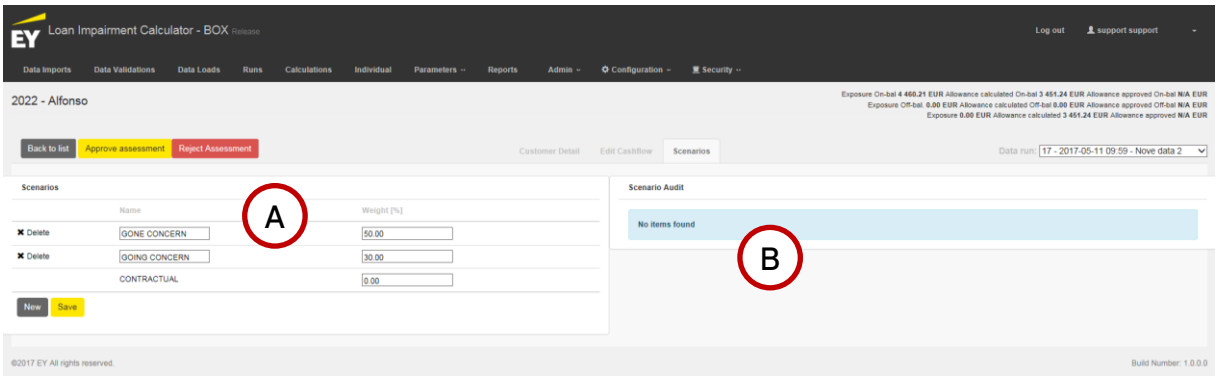

## 4.3.3 Scenario management

Each client has a CONTRACTUAL scenario, in which it is expected the client will pay as agreed in the contractual payment schedule. In such case, the calculated provision for the scenario is zero. By setting the probability of this scenario to 100%, zero provision is calculated. Cash flow cannot be assessed in this scenario. The individual provision is calculated as a weighted average of the scenario provisions. The formula is as follows:

$$Provision = \sum P_i * w_i$$

Where $w_i$ is the weight of scenario *i* and $P_i$ is the provision calculated for scenario *i*.

The *Scenarios* tab opens a screen section with an option to change scenario weights (A). The audit log (B) of the past scenario changes is also displayed.




**Point A – Scenario Manager**

In this part, the user can manage scenarios at the client level. In case of client-specific scenarios, the user can apply or change weights of individual scenarios by filling new values into the *Weight [%]* column. The user is also able to create new scenarios, or delete the existing ones (except for the *Contractual* scenario). The sum of the new percentages must equal to 100%. New values are saved once the *Save edits* button is pressed.

Default scenario settings to be applied to all individually assessed clients can be set in *Parameters* > *Global Scenario* described in detail in section 5.6.

**Point B – Log of Scenario edits**

Log of Scenario edits will be implemented in one of the next releases.

## 4.3.4 Calculation of individual allowance

Individual provision is calculated on the basis of cash flows and allocations, which were input in the tables described above.

Provision on an account is calculated using the account cash flow, customer cash flow and collateral cash flow. The last assessed cash flow (having payments in the future) of each type is used.

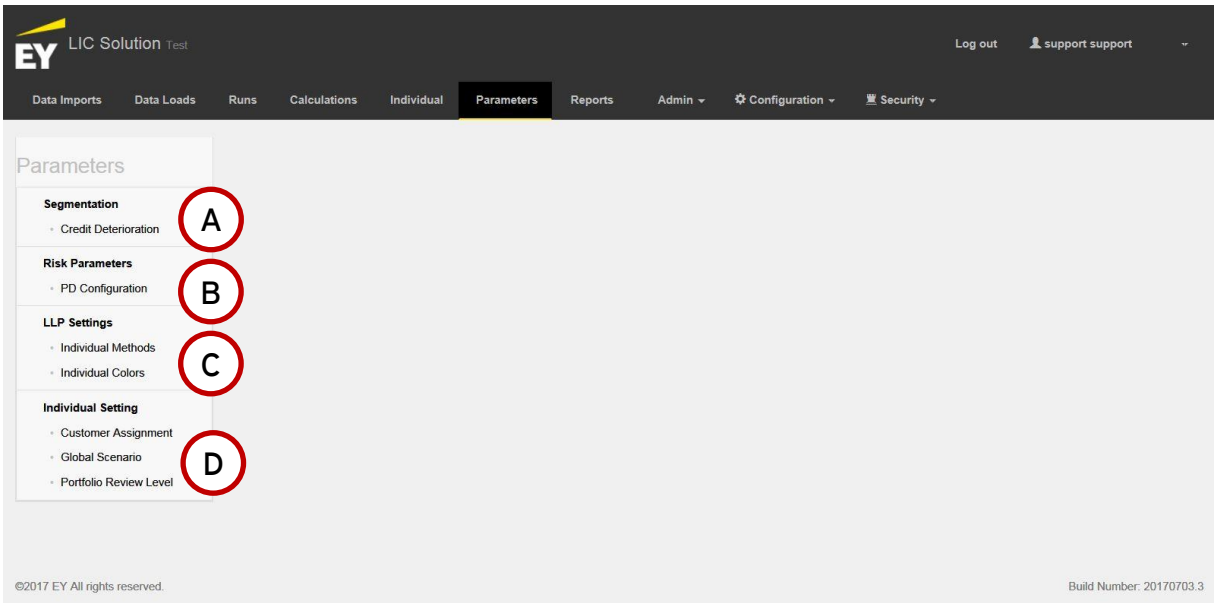
If the account whose provision is calculated has, for example, only cash flow on collateral and the cash flow was assessed and input in the past, only this cash flow is used for calculation of provision.

The calculation of reserves for off-balance exposures follows the same logic as the calculation of provisions. The level of reserve is determined by the difference between the account's off-balance exposure and the expected cash flows assigned to the account.

The *Submit assessment* button calculates the allowance, saves it and marks the current user of the module as the one that assessed the client.

# 5 Parameters

The *Parameters* section allows the user to set up the parameters for the calculation. Parameters affect the methods and conditions used for provision calculation.




Available parameters:

- *Credit Deterioration*
  - determines the rules for assigning stage to a client
- *PD Configuration*
  - determines methods for PD calculation
- *Individual Methods*
  - determines the rules for automatic selection of clients for individual assessment
- *Individual Colors*
  - determines the rules for review prioritization (red, orange and green) for clients in individual assessment
- *Customer Assignment*
  - Assigns individually assessed clients to the particular user
- *Global Scenario*
  - Determines default scenarios and probability split among scenarios for individual assessment
- *Portfolio Review Level*
  - Determines the parameters for a multi-level review process for each portfolio reviewed

## 5.1 Credit Deterioration

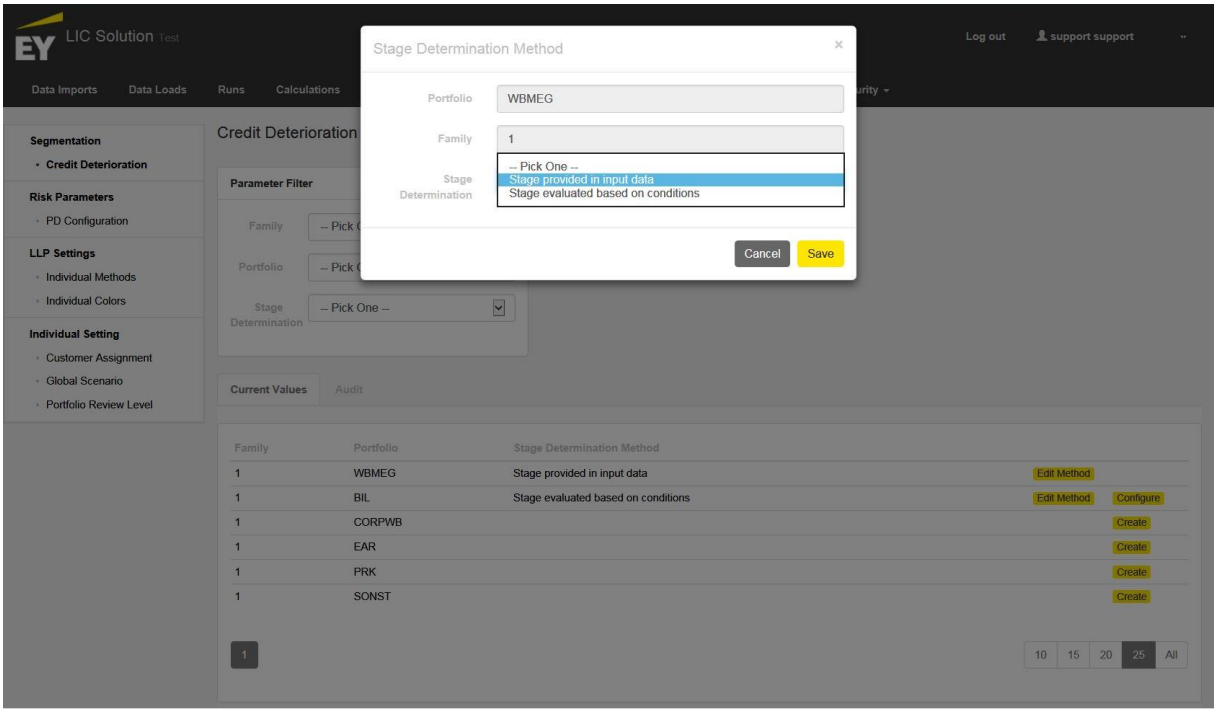
The *Credit Deterioration* section is accessible in the *Parameters* menu marked as *(A)* in the picture at the beginning of chapter 5.

For each portfolio, *Stage Determination Methods* must be selected from the following options:

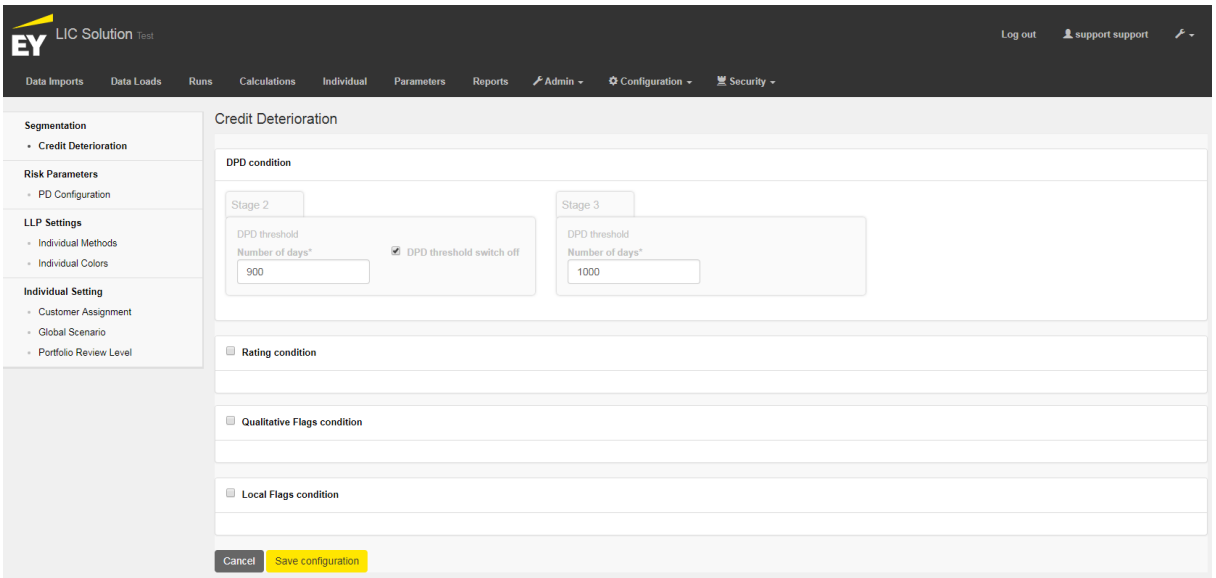
- *Stage provided in input data*
  - Stage of each account (client) defined in input tables.

- *Stage evaluated based on conditions*
  - Stage of each account (client) assigned by LIC based on specified conditions.

To select the *Stage Determination Method*, click on *Edit Method* button for the relevant portfolio, select one of the methods and click on the *Save* button:




If *Stage evaluated based on conditions* option is selected it should be further configured.
Click on the *Configure* button for the relevant portfolio and specify conditions through the interface:




In the current version, there are 4 types of conditions available:

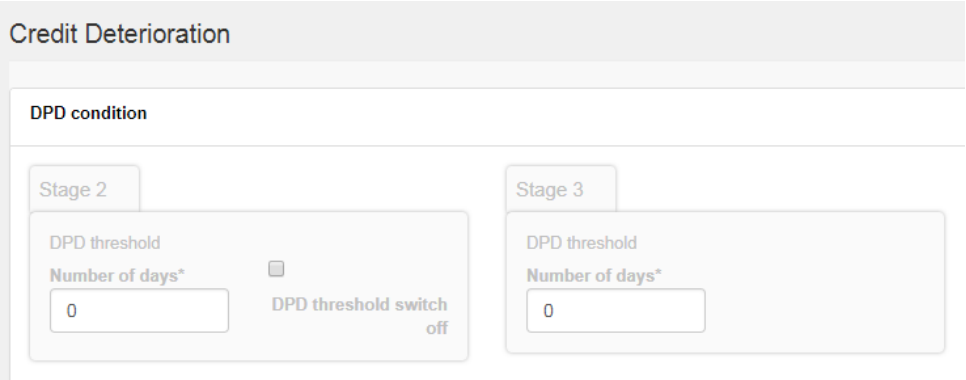
- *Days Past Due (DPD) condition*
  - determines DPD thresholds for accounts to be classified into Stage 2 and Stage 3

- *Rating condition*
  - determines thresholds for absolute rating condition for accounts to be classified into Stages and rating change condition for accounts to be classified into Stage 1 and Stage 2
- *Qualitative Flags condition*
  - determines the flags used for classification into Stage 2 and Stage 3.
- *Local Flags condition*
  - determines the local flags used for classification into Stage 2 and Stage 3.

The conditions are set up in such way that any condition triggering a higher Stage dominates the others.

### Days Past Due (DPD) condition

The user is asked to fill out two thresholds: the first one for evaluation between Stage 1 and Stage 2 (field *Stage 2 – Number of days*) and the second one between Stage 2 and Stage 3 (field *Stage 3 – Number of days*). The DPD value below the threshold *Stage 2 – Number of* days indicates Stage 1, the value of DPD above this threshold but below threshold *Stage 3 – Number of* days indicates Stage 2, and the value above threshold *Stage 3 – Number of days* indicates Stage 3.

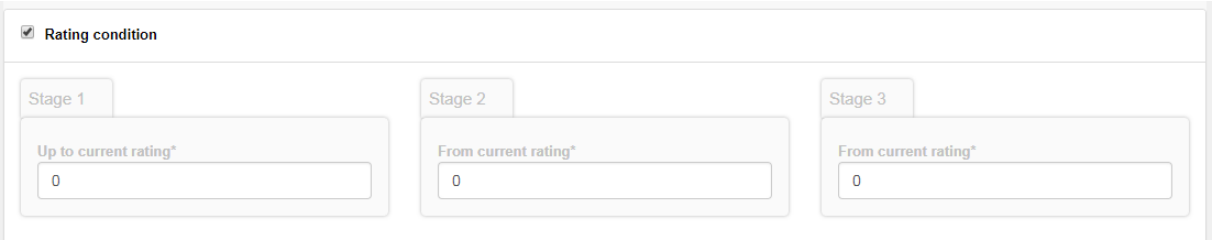



### Rating condition

<u>Absolute rating condition:</u>

1. The user is asked to fill out the *Stage 1 - Up to current rating* field that represents the worst rating for which a client still satisfies the absolute rating condition for Stage 1.
2. Then the user fills out the *Stage 2 – From current rating* field, representing the best rating for which the absolute rating condition for Stage 2 is satisfied.
3. Similarly, the *Stage 3 – From current rating* field determines the best rating for a client for which the absolute rating condition for Stage 3 is satisfied.

All fields mentioned in points 1–3 above are filled by an integer from the *Hierarchy* row.




<u>Change in rating:</u>

The field *Origination rating threshold* represents the threshold triggering Stage 2 by the criterion of rating change as opposed to the criterion of absolute value of current rating.

The user is asked to fill *Origination rating threshold* for each value of current rating in the interval of [“*Stage 1 - Up to current rating*” + 1; “*Stage 2 – From current rating*” – 1] (i.e. for all ratings which are outside of the rating intervals assessed by the absolute rating conditions). This threshold represents the worst client rating as at origination for which the change to current rating is considered as significant deterioration, i.e. Stage 2 is assigned.

To set the rating threshold values the user fills in the integer values from the *Hierarchy* row noting that only values lower than the *Hierarchy* values for *current rating* are admissible.

Stage 1 - 2 from rating evaluation

| | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Hierarchy | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 2 | 3 | 4 | 5 | 6 |
| Current rating | 1 | 2 | 3 | 4a | 4c | 5a | 6b | 8 | A | A2 | B1 | B2 | C2 | D | D1 | D2 | N | R | R1 | R2 | R3 | R4 | R5 |
| Origination rating threshold | | | | | | | | | | | | | | | | | | | | | | | |

Staging Origination Rating Method

Stage 2 assigned

### Qualitative Flags condition

1. The user is asked to check the flags which are to be used for the stage evaluation (checklist *Active*).
2. Then the user chooses the stage a particular flag triggers (either Stage 2 or Stage 3 is selected in the *Stage* column).
3. Finally, the user has an option to check the field  for the *Stage 2 switch off:*
   - If the *Stage 2 switch off* is checked, then, if the current flag is 1, and it was 1at the date of origination, then the flag condition is evaluated as Stage 1. If the current flag is 1 and at the date of origination it was 0, then the flag condition is evaluated as Stage 2.
   - If the *Stage 2 switch off* is not checked, then the current flag value of 1 is evaluated as Stage 2 regardless of the flag value at origination

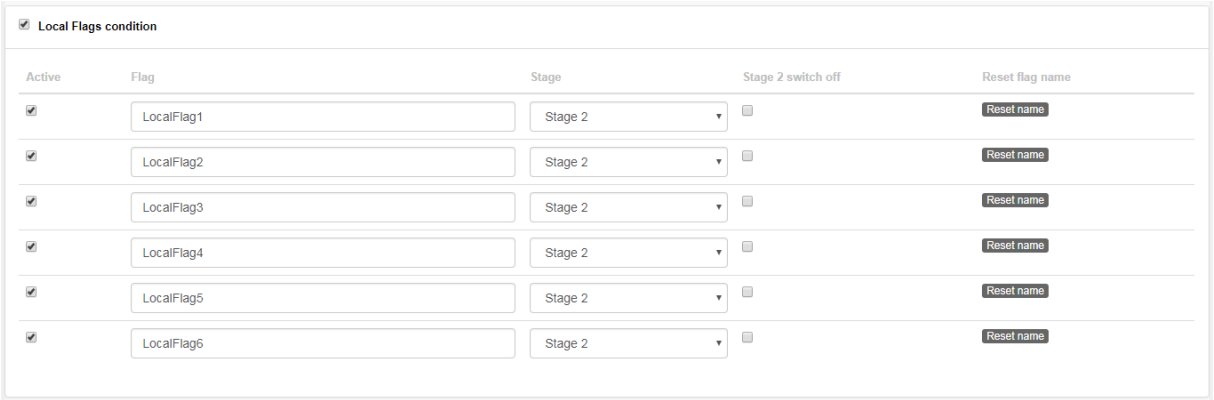
Qualitative Flags condition

| Active | Flag | Stage | Stage 2 switch off |
|---|---|---|---|
| ☑ | Default flag | 3 | |
| ☑ | Watchlist flag | Stage 2 | ☐ |
| ☑ | Insolvency flag | Stage 2 | ☐ |
| ☑ | Default in GCC Clients flag | Stage 2 | ☐ |

### Local Flags condition

1. The user is asked to check the flags which are to be used for the stage evaluation (checklist *Active*).
2. Then the user chooses the stage a particular flag triggers (either Stage 2 or Stage 3 is selected in the *Stage* column).
3. Finally, the user has an option to check the field  for the *Stage 2 switch off:*
   - If the *Stage 2 switch off* is checked, then, if the current flag is 1, and it was 1at the date of origination, then the flag condition is evaluated as Stage 1. If the current flag is 1 and at the date of origination it was 0, then the flag condition is evaluated as Stage 2.
   - If the *Stage 2 switch off* is not checked, then the current flag value of 1 is evaluated as Stage 2 regardless of the flag value at origination
4. The application enables the user to rename the local flags by rewriting their name and clicking on the *Save configuration* button. By clicking on the *Reset name* button, the user is able to reset the

name of the local flag to its original value. This change also needs to be confirmed by the *Save configuration* button.




Each condition may be enabled or disabled independently. After configuring the required conditions click on the *Save configuration* button.

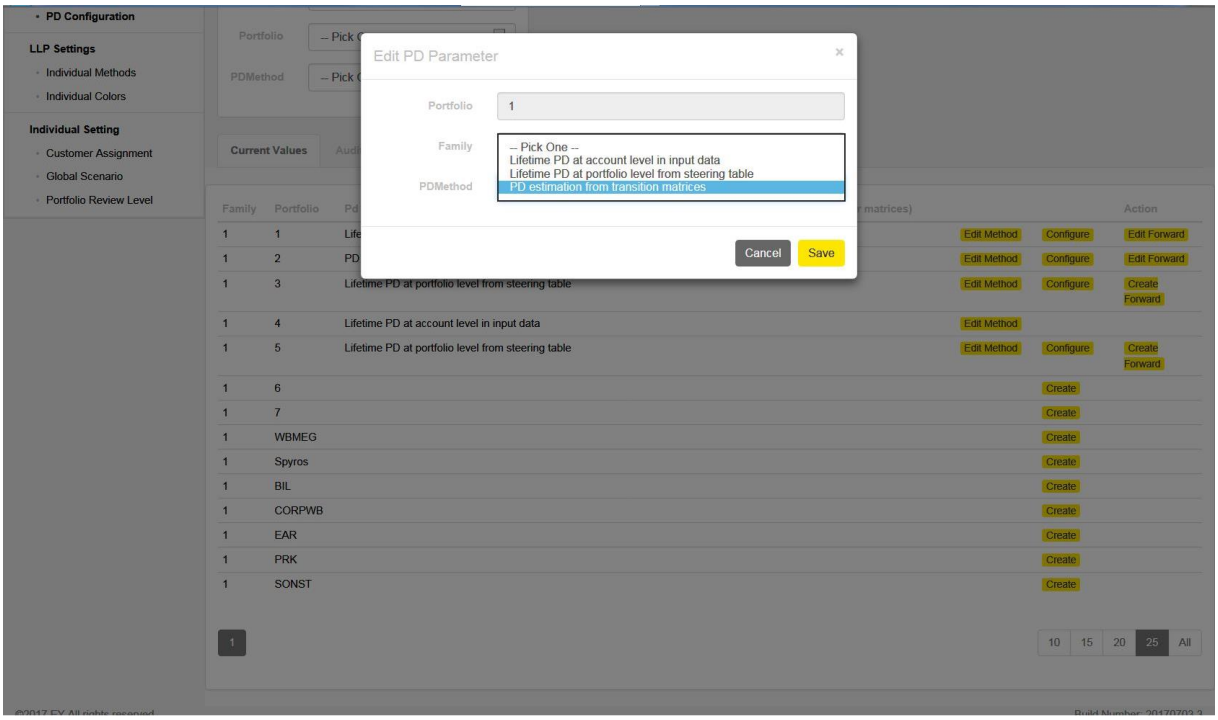
## 5.2 PD Configuration

The section *PD Configuration* is accessible in the *Parameters* menu, marked as *(B)* in the picture at the beginning of chapter 5.

For each portfolio, a PD method must be selected from the following options:

- *Lifetime PD at account level in the input data*
  - PDs are provided in the input data for each account
- *Lifetime PD at portfolio level from steering table*
  - PDs are provided in the input data for each portfolio
- *PD estimation from transition matrices*
  - PDs are calculated by LIC based on historical data. **Note:** this method will be implemented in one of the future releases.

To select a PD method, click on the *Create* or *Edit Method* button for the relevant portfolio, select one of the methods from the list and then click on the *Save* button:



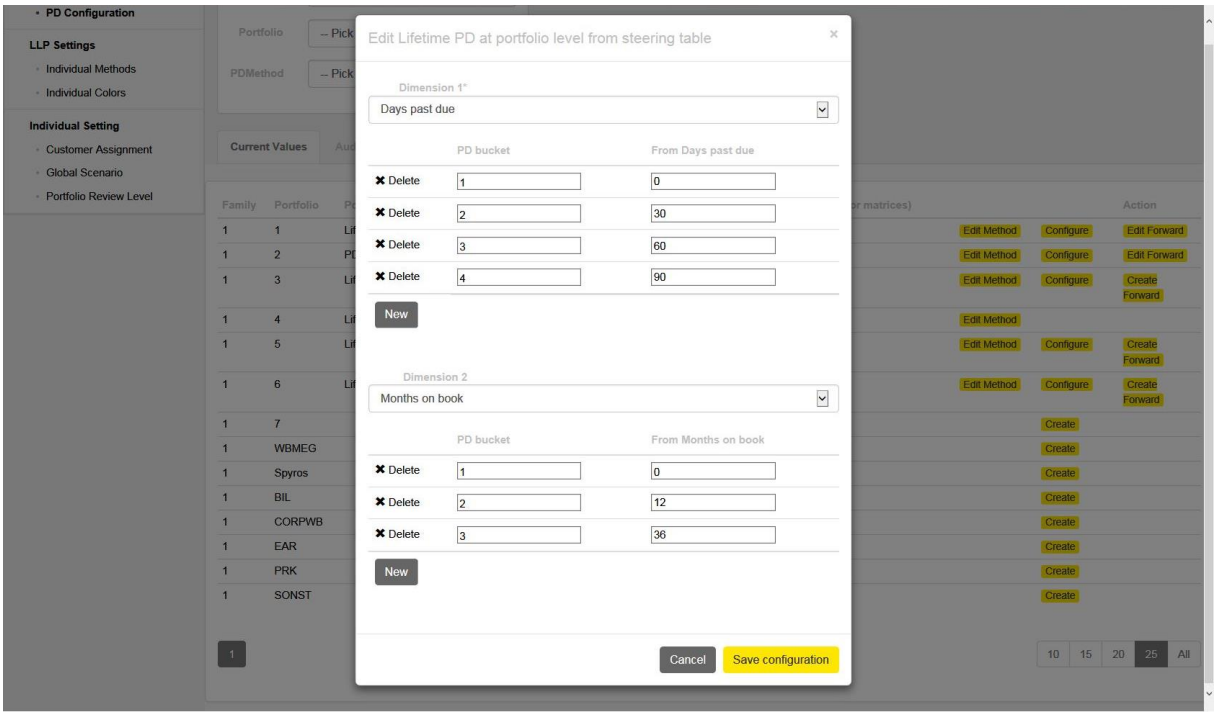

If *Lifetime PD at account level in the input data* option was selected, no further configuration is required.

For *Lifetime PD at portfolio level from steering table* and *PD estimation from transition matrices* options, parameters must be configured.

***Lifetime PD at portfolio level from steering table*** **configuration:**

There are up to 2 dimensions that can be configured for PDs. The first dimension can be either *DPD buckets* or *Rating buckets*. The second dimension can be either *Loan to Value (LTV) ratio* or *Months on book (MOB)*. Please note that the second dimension can be used only after the first dimension is configured.

PDs for each combination of portfolios and up to 2 dimensions are provided in the input data.




# 5.3 Individual Methods

The section *Individual Methods* is accessible under the *LLP Settings* of the *Parameters* menu marked as *(C)* in the picture at the beginning of chapter 5.

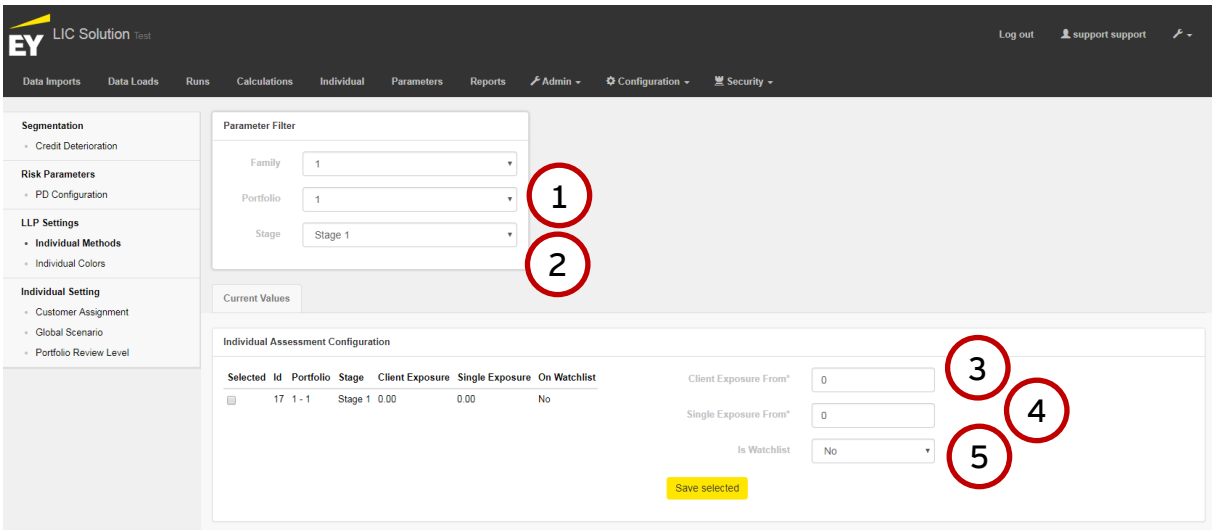
Conditions for automatic selection of the client for individual assessment are defined in this section.

Conditions for individual assessment could be configured combining the following parameters of the client:

- *Portfolio (1)*
- *Stage (2)*
- *Client Exposure From (3)*
- *Single Exposure From (4)*
- *Watchlist flag (5)*

Conditions within one line are united by an *AND* operator. This means that all conditions within one entry must be true for an account to be assigned to individual assessment.

Each line of conditions is considered separately (united by an *OR* operator).




# 5.4 Individual Colors

The section *Individual Colors* is accessible under the *LLP Settings* of the *Parameters* menu marked as *(C)* in the picture at the beginning of chapter 5.

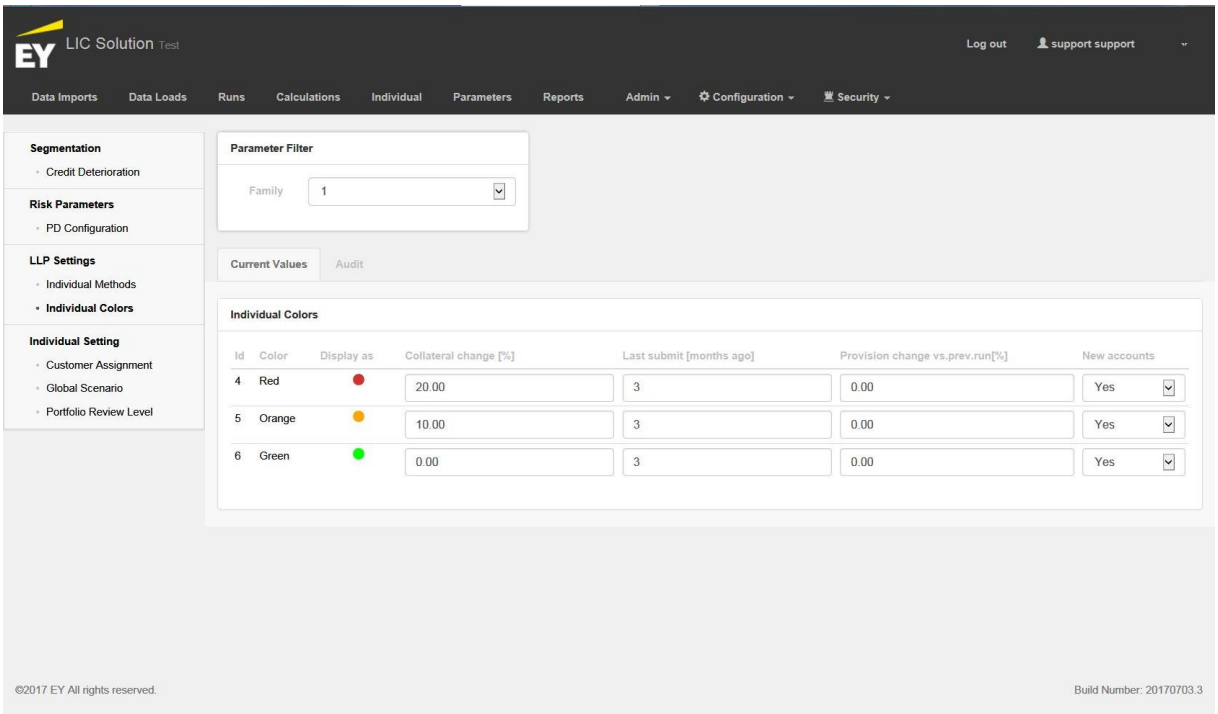
Conditions for client review prioritization (Red, Orange, Green) should be set up in this section.

Conditions are based on changes compared to previous run. Clients with detected higher relative changes should have higher review priority. In addition to this, clients with new accounts and clients with long intervals from the last individual assessment are generally also prioritized for the assessment.

These conditions are applied to individually assessed clients and include:

- *Collateral change (%)*
- *Last submit (months ago)*
- *Provision change vs. previous run (%)*
- *New account flag*

The values of criteria represent the lower bounds for the interval at which the client is assigned by Priority Color. The conditions are connected by an OR logical relationship.

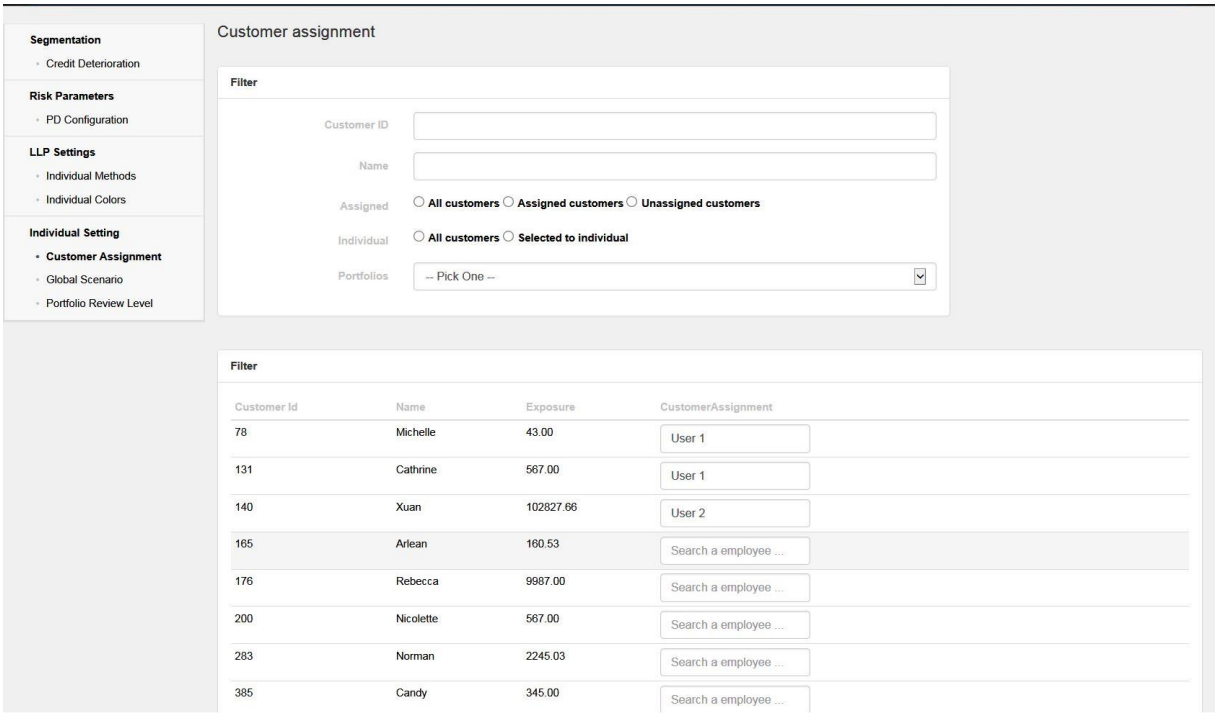



# 5.5 Customer Assignment

The section *Customer Assignment* is accessible under the *Individual Settings* of the *Parameters* menu highlighted as *(D)* in the picture at the beginning of chapter 5.

Individually assessed clients may be assigned to a particular user.

This parameter is used as an additional filter in the individual assessment section: individually assessed clients may be filtered by an assigned user.



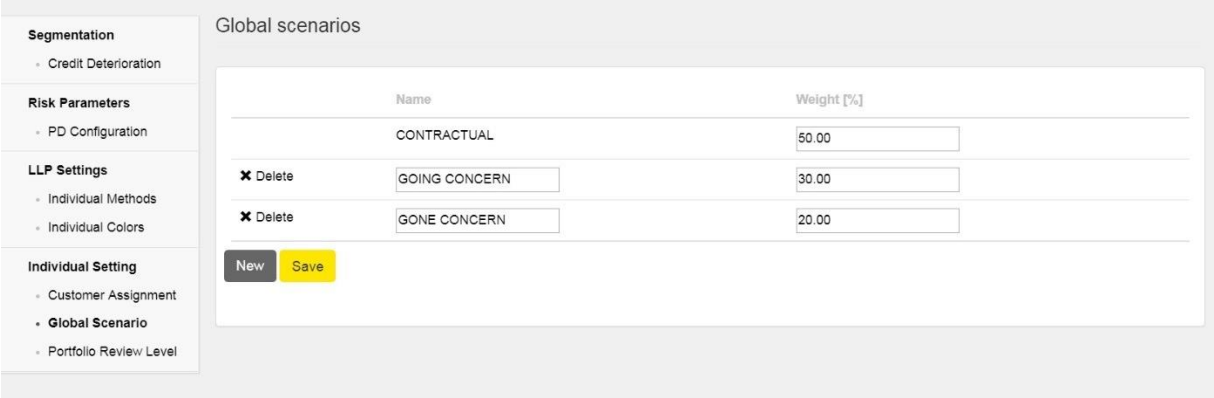
# 5.6 Global Scenario

The section *Global Scenario* is accessible under the *Individual Settings* of the *Parameters* menu marked as *(D)* in the picture at the beginning of chapter 5.

*Global Scenario* opens a section where the user has an option to manage scenarios globally, i.e. for all clients. If a previous run exists, the settings from this run are automatically applied to the new run as well. Edited *Global Scenario* settings will only be applied to runs that are not tied to any previous runs.

Up to 5 different scenarios may be created. Users can also specify predefined (global) probabilities for each scenario, which may be changed later for each particular client during individual assessment.

All edits must be saved. Global parameters do not effect clients with scenario parameters changed manually.



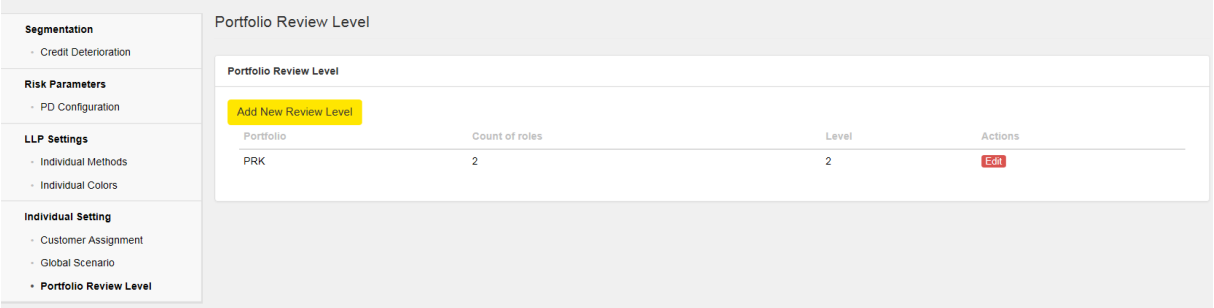

# 5.7 Portfolio Review Level

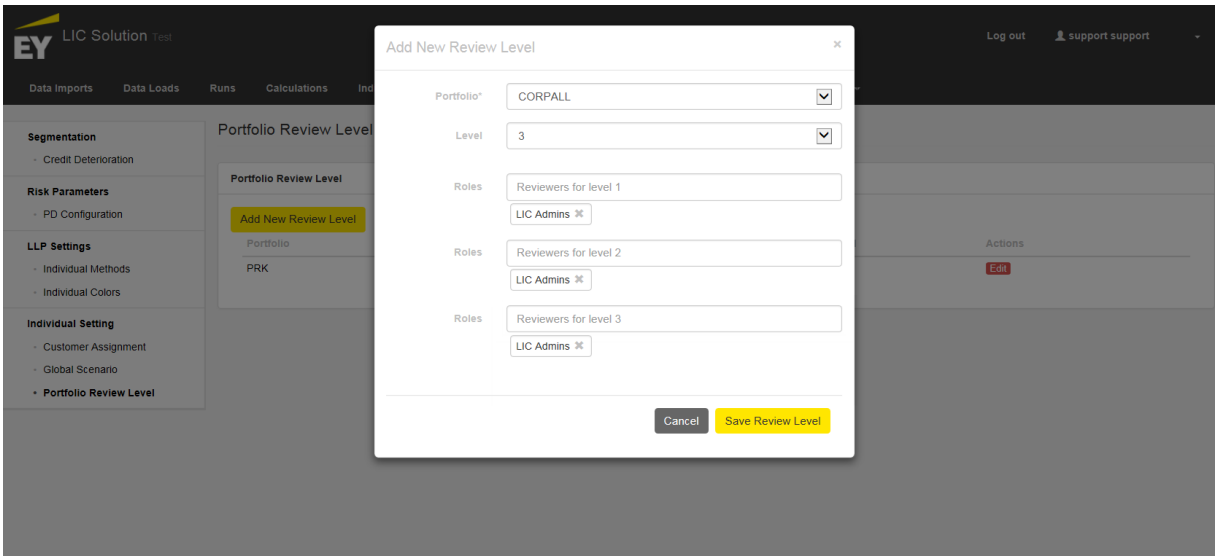
The section *Portfolio Review Level* is accessible under the *Individual Settings* of the *Parameters* menu marked as *(D)* in the picture at the beginning of chapter 5.

In the *Portfolio Review Level* section the user defines the level of multi-level review for each portfolio and sets the reviewers for each specific review level.

New review levels for portfolios are added using the *Add New Review Level* button. The modification and removal of existing review levels is possible using the *Edit* button.

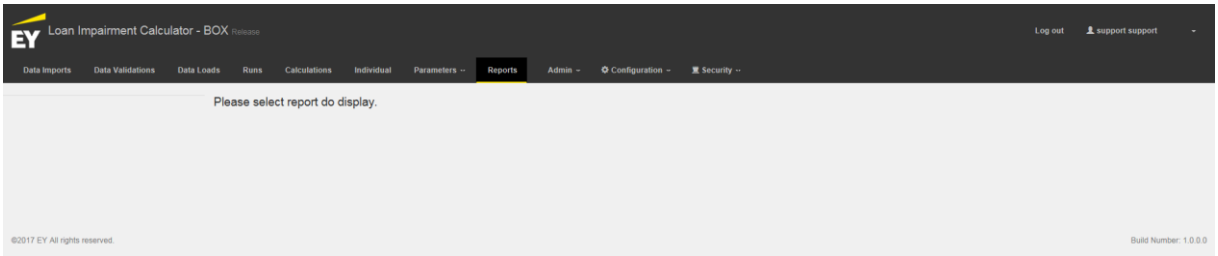
The screens below display the related user environment.




LIC Solution Test
Log out
support support
Data Imports
Data Loads
Runs
Calculations
Segmentation
Credit Deterioration
Risk Parameters
PD Configuration
LLP Settings
Individual Methods
Individual Colors
Individual Setting
Customer Assignment
Global Scenario
Portfolio Review Level
Portfolio Review Level
Portfolio Review Level
Add New Review Level
Portfolio
PRK
Actions
Edit
Add New Review Level
Portfolio*
CORPALL
Level
3
Roles
Reviewers for level 1
LIC Admins
Roles
Reviewers for level 2
LIC Admins
Roles
Reviewers for level 3
LIC Admins
Cancel
Save Review Level

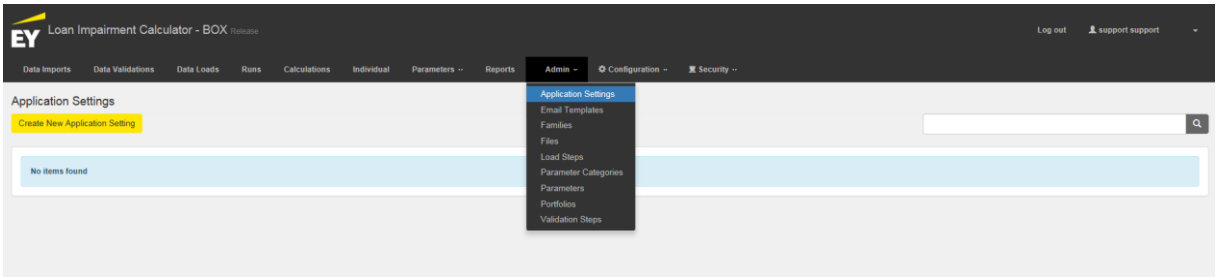
# 6 Reports

This section is under construction and will be updated in further releases.



# 7 Admin

This section is under construction and will be updated in further releases.

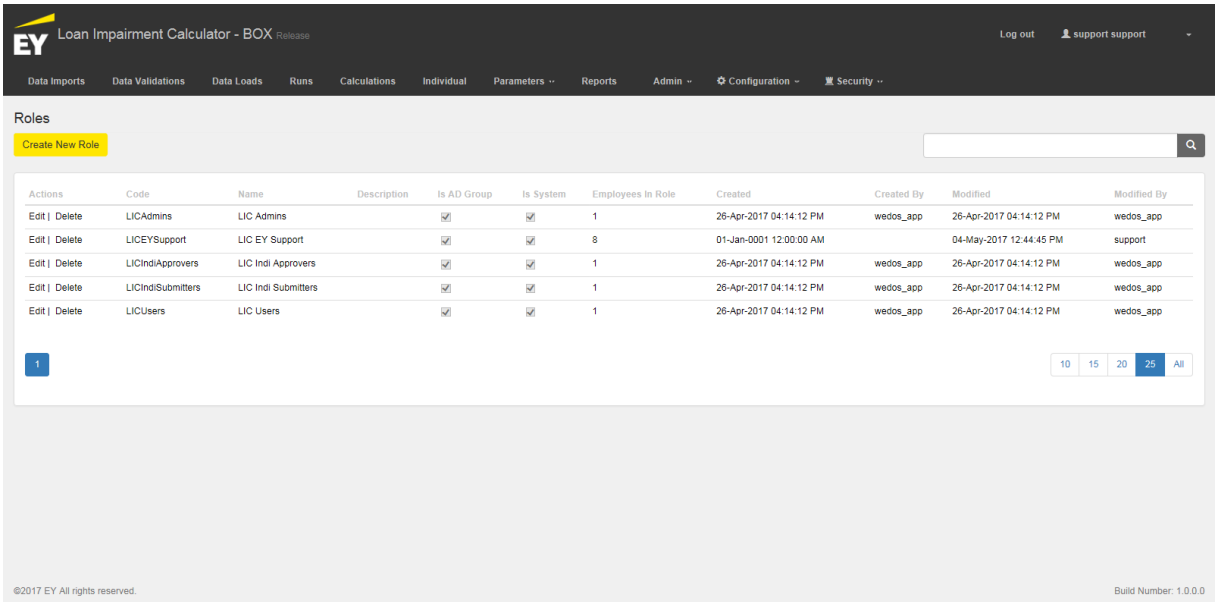


# 8 Security

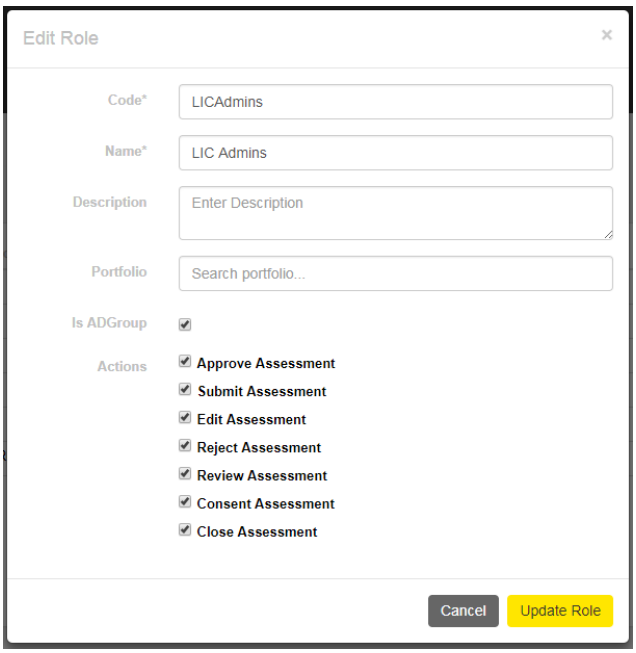
In the *Security* section, access rights are managed. An authorized user may set up new or adjust existing user roles for individual users.

## 8.1 Roles

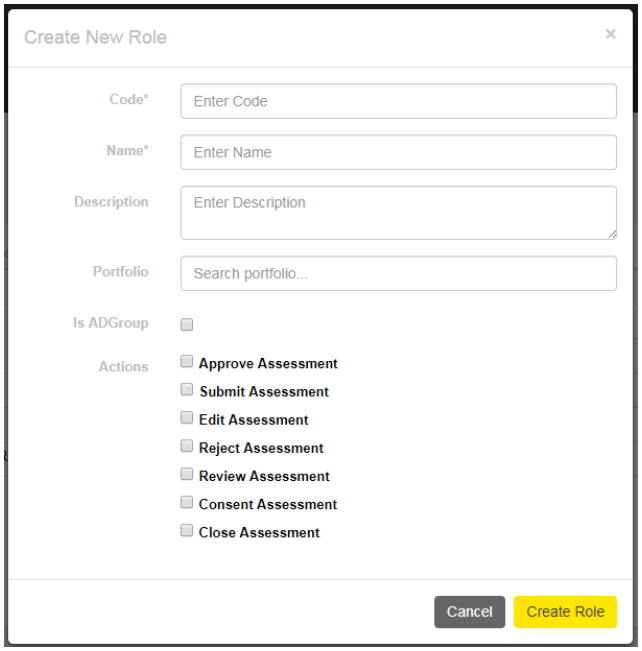
User roles can be edited in the following screen




The screen shows a list of all existing roles. By clicking on the *Edit* button, the properties of the role can be altered by checking a particular box from the list of available options/actions. Any changes made to the role must be confirmed by pressing the *Update role* button.



After pressing the *Create new role* button, a pop-up window appears. In this window, a user can create a new role. *Code* and *Name* fields are mandatory while description and other fields are optional. If the user wants to create a role that is mapped to an Active Directory (AD) Group, he needs to select the *Is AD Group* checkbox. *Is System* checkbox is used for roles that are not synchronized with AD and should not be deleted. Creation of a new role is confirmed by pressing the *Create Role* button.




## 8.2 Active directory

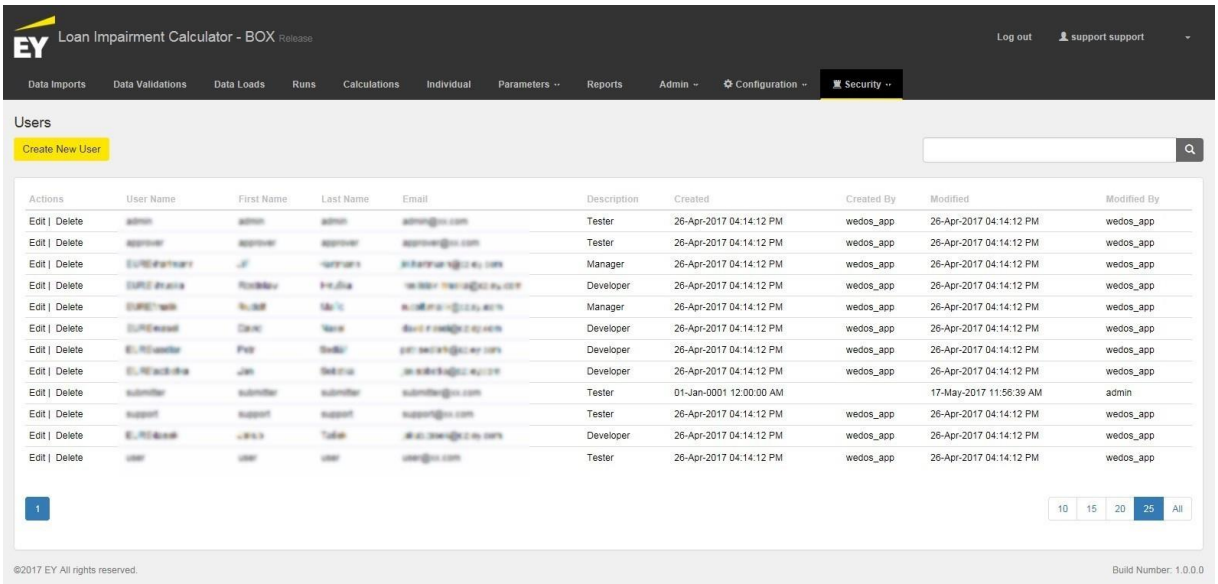
For easier user management, it is possible to use LDAP groups for handling user authentication and authorization.

In order to use this feature, it is necessary to create one or more roles in LIC, where the code of the role is equal to the LDAP group name. Please note that *Is AD Group* option must be selected when creating the role. If a user is a member of a LDAP group that exists in the LIC roles, then the user profile is created automatically in LIC and his groups are automatically synchronized with the LIC roles.

It is recommended to create two roles: one role with a permission to manage *Security*, and a second role for normal application users without permissions to *Security*, e.g.: LIC Users and LIC Admins.

## 8.3 Users

The user settings can be added or edited in the screen displayed below.

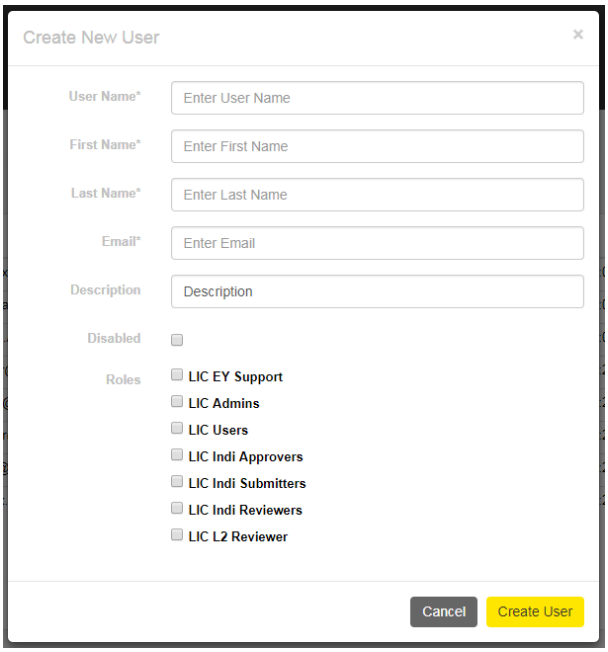



When one of the users is selected by clicking on *Edit*, user credentials appear in a new pop-up window. It is possible to edit *user name*, *first name*, *last name*, *email address* and *roles*.

The following steps should be followed to create or update user details:

- Click on the *Create New User* button
- Add *User Name*, *First Name*, *Last Name*, *Email*, *Roles* and *Description*[1]
- If the user should not be able access the application at the moment, check the box next to *Disabled*. This selection can be later changed by editing the existing user.
- Submit by clicking on the *Create User* button

Role of the user can be selected from the list next to the *Roles* headline.




<***Note:*** *It is recommended not to create new roles but to use the existing set.*>

[1] Fields marked with * are mandatory. *User Name* can only contain letters a-z, A-Z, numbers 0-9 and special characters (!,@,#,\etc.) with no spaces in between the characters

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